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Chapter 7: Fermi Estimation of the Cognitive Reynolds Number—Information-Arrival Pressure, Verification Capacity, and Debt Trajectories

Abstract

Taking the history of media change as its axis, this chapter traces a single thread: how the power controlling the media conductivity coefficient shifted from the host to the projecting party, and why partial restitutions of control have not amounted to a systematic reversal; it then uses a symbiosis-parasitism continuum to explain the evolutionary dynamics behind this migration.

The *quasi-cognitive Reynolds number* (Re_c; the full term marks the distinction from the physical Reynolds number, while the running text uses **cognitive Reynolds number**) is not a physical quantity that can be precisely calculated, but it is an effective Fermi comparison tool. Starting from the general form in Chapter 3, this chapter first specifies the conditions under which the simplified form $Re_c \approx \Lambda / E_c$ applies to same-scenario comparisons over short time slices, and then constructs scenario-specific Fermi estimates in audit time; in the simplified form, it compares information-arrival pressure with cognitive processing capacity: the numerator is the media conductivity coefficient (Λ), the denominator the human biological substrate (E_c, approximately 10 bits/s—a representative order-of-magnitude summary given by the Zheng and Meister (2025) review, approximately constant on the time scale of civilizational history).

This chapter estimates their ratio by the within-scenario normalization of Chapter 3, with both sides reckoned in the time calibration requires: the numerator is the time required to audit the information arriving per unit time

in a scenario that the host must verify; the denominator is the length of the arrival stream itself, with allocated verification time recorded separately; 1 is the capacity boundary of that scenario. In books the reader sets the pace and can press the audit-awaiting arrival below the verification capacity; subtitle speeds above roughly 20 characters per second—the rate of silent reading—lead viewers to shift toward coarser skimming; the audit-awaiting arrival of short videos and generative artificial intelligence (AI) grows with item counts and generation volume, while calibration capacity is constrained by the human biological substrate and does not grow with technology. The audio-track reading component of short videos is about 0.63–0.84, with the overload pressure carried by the compression–restoration verification component and the deferability criterion (under explicitly stated scenarios the verification component reaches severalfold to tens of times, depending on content type and responsibility structure); the chapter's AI demonstration is about 5.6 (the reading component at the midpoint of the generation-volume range). These figures are scenario estimates under explicitly stated parameters (§2.3). Cross-media comparison concerns each scenario's ratio relative to its capacity boundary; no medium's arrival rate is written as a multiple of the human biological substrate (E_c); under the same host and the same reckoning, order-of-magnitude comparison is admissible (§2.2, §6.2).

Keywords

cognitive Reynolds number; media conductivity coefficient; human biological substrate; Fermi estimation; laminar-turbulent transition; history of media change; stop signal; symbiosis-parasitism continuum; symbiosis reversal; verification window

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§1 Problem Formulation and Epistemological Constraints

§1.1 Media as the Historical Carrier of the Media Conductivity Coefficient (Λ)

The formal derivation in Chapter 3, §5.2 presents the general form of the quasi-cognitive Reynolds number before introducing its simplified form for same-scenario comparisons over short time slices. The general form is:

$$Re_c = \frac{C \cdot \Lambda \cdot \phi}{A \cdot E_c \cdot \|L\|}$$

The projection side comprises unlocked stock (C), the media conductivity coefficient (Λ), and narrative load (ϕ); the defense side comprises meta-audit capacity (A), the human biological substrate (E_c), and the legitimacy base ($\|L\|$). This expression identifies the structural relation between total projection-side load and total defense-side capacity; it is not used directly for numerical calculation.

For same-scenario comparisons by an individual over short slices of months to years, when unlocked stock, narrative load, meta-audit capacity, and the legitimacy base can be treated as background conditions and absorbed into the scenario scale, the heuristic simplified form is:

$$Re_c \approx \frac{\Lambda}{E_c}$$

When meta-audit capacity must be tracked separately, Chapter 3, §5.3 expands the denominator into cognitive friction ($\mu_c = A \cdot E_c$), giving $Re_c \approx \Lambda/\mu_c$. The first three media intervals of this chapter use the two-variable simplified form; the AI scenario returns to the general form. Simplification is a scenario-specific operation, not the permanent deletion of the remaining variables.

Innis (1951) systematically revealed the deep shaping of civilizational structures by the material form of media: stone carvings favor time (durable but difficult to transmit), papyrus favors space (lightweight but perishable)—the physical properties of media are themselves mechanisms for the distribution of power, prior to all content. McLuhan (1964) made a more radical judgment—"the medium is the message": the true transformation of human society does not come from the content carried by the medium but from the reshaping of cognitive modes by the medium form itself.

In the Macro-Cognitive Accounting Framework (MCAF), the media conductivity coefficient (Λ) **abstracts an operationalizable dimension from the insights of Innis and McLuhan**—"information flow rate and cognitive load"—rather than fully reproducing the totality of their civilizational diagnoses. The media conductivity coefficient not only measures the rate at which information arrives but also encodes the structural pressure imposed on the host by the medium form. When the medium form evolves from books to AI-generated content, what changes is not merely the quantity of content but the power relationship itself between the host and information.[1]

However, a crucial connection must be stated before the concept is put to use. The thousand-clocks intuition of Chapter 1, §1.4 and the boundary propositions of §3.5 jointly establish: an increase in the number of mappings yields no information gain about reality—a thousand uncalibrated clocks are epistemologically equivalent to one uncalibrated clock (the proposition holds under the boundary condition that the mapping functions are entirely unknown—see Chapter 1, §3.5). This proposition directly constrains the analysis of this chapter: a higher media conductivity coefficient means more readings have arrived at the host; it does not mean that the fiber between the readings and reality has shrunk. More clocks do not equal more accurate time. The rise of the cognitive Reynolds number (Re_c) measures the pressure of the information flow, not the quality of the information.

§1.2 The Boundary Between the Physical Reynolds Number and the Quasi-Cognitive Reynolds Number

In 1883, Reynolds (1883), through pipe flow experiments, discovered that fluid motion exists in two states, laminar and turbulent, with the transition (the standard fluid-mechanics term for the process by which laminar flow changes into turbulent flow) determined by the Reynolds number:

$$Re = \frac{\rho v L}{\mu}$$

The physical meaning is the ratio of inertial force to viscous force. A standard reference value in modern fluid mechanics for circular pipe flow: the critical Reynolds number is approximately 2100 (for circular pipe flow; the critical value differs for different geometries)—a reference threshold that can be experimentally reproduced

[1] Mapping the media theories of Innis and McLuhan onto a single dimension of the media conductivity coefficient is a **drastic simplification** by the framework of this chapter. The "time bias/space bias" emphasized by Innis and McLuhan's "the medium is the message" contain a holistic diagnosis of the cognitive modes of civilization, far beyond the scope of "information arrival rate." Specifically, McLuhan defines a medium's message as the change of scale, pace, or pattern that the medium or technology introduces into human affairs (McLuhan, 1964); the railway's message lies precisely in its acceleration and enlargement of pre-existing human functions. This chapter's extraction of the information-flow-rate dimension is an **analytical cut** of that proposition, not another question orthogonal to it; the "cognitive mode reshaping" proposition cut away here requires analytical tools different from the information rate and lies outside the domain of this chapter.

(its value depends on disturbance conditions, and standard values in different textbooks vary slightly between 2000 and 2300; see below) (Munson et al., 2013). Both 2100 and the 2100–4000 transition zone mentioned below are standard values of modern fluid mechanics. Reynolds's 1883 paper speaks in terms of the "critical velocity," and records two distinct critical velocities—the one at which steady motion broke down, and the one on which all his experiments agreed, the former being much greater than the latter: the two critical velocities are proportional to 1/43.7 and 1/278 respectively, the breakdown velocity being roughly 6.4 times the concordant critical velocity ($278 : 43.7 \approx 6.36 : 1$)—while noting that the critical velocity is highly sensitive to disturbances at the pipe inlet; the paper gives no such dimensionless number.

What the cognitive Reynolds number (Re_c) borrows is the structural intuition—the ratio of information inertia (media conductivity coefficient) to cognitive viscosity (human biological substrate)—not the physical quantity itself. Three fundamental differences must be clarified before use:

- **Mathematical Form and Dimensionality**: The physical Reynolds number (Re) is a strictly dimensionless number. The numerator of the cognitive Reynolds number, the media conductivity coefficient, is a multi-dimensional feature set (§2.2) that cannot, as a whole, be reduced to a single scalar; the ratio in the numerical contexts of this chapter is defined only within specific scenarios: both sides are reckoned in the time calibration requires—the numerator is the time required to audit the information arriving per unit time in that scenario that the host must verify; the denominator is the length of the arrival stream itself, with allocated verification time recorded separately (the "information arrival / auditable amount" of Chapter 3, §5.2). Dividing quantities in the same unit yields a pure number in mathematical form; what must be distinguished is the comparability boundary of that pure number—it is the result of within-scenario normalization, 1 is the capacity boundary of that scenario, and it holds no cross-system status as a physical similarity constant.
- **Transition Criteria**: The physical Reynolds number has an experimentally investigable system of transition criteria, but the critical value depends on geometry and disturbance conditions (2100 is the standard reference value for circular pipe flow, and 2100–4000 is the transition zone)—it is not a universal exact constant. The cognitive Reynolds number has no corresponding experimentally reproducible criterion, only an ordinally distinguishable transition zone.
- **Operationalizability**: The variables of the physical Reynolds number can all be directly measured. The numerator of the cognitive Reynolds number depends on order-of-magnitude judgments from proxy indicators.

The cognitive Reynolds number is a heuristic tool for ordinal comparison: it is used to judge whether a scenario's ratio lies below, near, or above that scenario's capacity boundary, and the direction of change—not to provide precise numerical values. The "borrowing" here follows the fundamental discipline of structural analogy—taking only the structural isomorphism, not making a 1:1 mapping; the critique by Sussmann and Zahler (1978) of the migration of Catastrophe Theory into the social sciences applies equally here. The implication of this declaration for the scope of application of what follows is: what the cognitive Reynolds number (Re_c) retains is the **ordinal comparison intuition** of the physical Reynolds number—changes in the order of magnitude of the ratio indicate the direction of possible state transitions of the system—rather than its precise critical value. Directional judgment is the valid inferential range of the cognitive Reynolds number (Re_c); all estimates of the cognitive Reynolds number (Re_c) for various media stages in this chapter are uniformly to be read under this ordinal framework and are not to be used as precise measurements. (Terminological note: "ordinal comparison"

as used in this chapter means comparing each scenario's ratio relative to its capacity boundary. A within-scenario ratio can be computed in the same units to a rough multiple, and quantities of the same construct can be compared in orders of magnitude; when the same host, the same task, the same information unit, and the same accounting period hold, and calibration capacity is identical or separately estimable, the ratios of two scenarios can also be compared directly, read in orders of magnitude (same order, one order higher), not as precise multiples; when these conditions are not met, the ratios do not form a single axis, and no cross-media assertion such as "higher by two orders of magnitude" is made.)

Order-of-magnitude comparison in this chapter follows three licensing conditions: **construct declaration**—what is compared is the audit-awaiting information and the auditable amount within a scenario, both reckoned in the time auditing requires, not sensory-layer bits and not encoded transport streams; each medium is reckoned in its default usage mode (who sets the rhythm is itself a difference in control; see §2.2). **Same units**—order-of-magnitude comparison is made only between quantities of the same unit and the same construct, for example subtitle characters per second against silent-reading characters per second, or average shot lengths across decades under the same counting convention; numbers obtained by different methods of measurement are neither added nor divided. **Standardized comparison**—1 is the capacity boundary of that scenario, not a cross-scenario constant (Chapter 3, §5.2); cross-scenario comparison requires first unifying the host, the task, the information unit, and the accounting period—when these conditions hold, comparison proceeds in orders of magnitude; otherwise, only the position and direction of each ratio are compared. Order-of-magnitude claims satisfying these three conditions are the strict form of being "vaguely right" (stated in full in §6.3): magnitude retained, modality honest. Claims failing them are pseudo-precision.

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§2 Denominator and Numerator

§2.1 The Human Biological Substrate (E_c): The Biological Constraint of Approximately 10 bits/s

The review published by Zheng and Meister (2025) in *Neuron* summarizes the information throughput of human conscious behavior across behavioral tasks as being on the order of 10 bits/s—the estimates listed in Table 1 of that review range from about 5 to 50 bits/s (reading 28–45, object recognition 30–50, listening comprehension about 13, choice-reaction about 5), all within one order of magnitude; 10 serves as the representative figure, not a ceiling.

This figure is extracted from a large number of different behavioral tasks—professional typing, normal reading, video game control, speech output, blindfolded speedcubing—no matter how the task type changes, the information rate of conscious decision-making always converges on the same order of magnitude. The rate at which sensory systems collect data is approximately 10^9 bits/s (all sensory channels combined; the visual-chain figures given in the same review are about 1.6 Gbit/s for the roughly 6 million cones of one eye, whereas the capacity of the optic nerve is about 100 Mbit/s or less—the retina has already compressed the visual signal by at least a factor of ten), but there exists a bottleneck of approximately 10 bits/s at the conscious processing stage. ("Bottleneck" here means an architectural constraint under evolutionary selection, not an engineering design ceiling; the candidate mechanisms—the statistical properties of neural coding and finite logical depth—are given in von Neumann's (1958/2012) early argument, on which Zheng and Meister (2025) themselves withhold judgment, and this chapter renders no verdict. Every later use of "evolutionary architectural constraint" takes

this sense.)

The bottleneck above specifically refers to **conscious, serially-attended information processing**. The processing capacity of unconscious parallel processing (such as rapid parsing of visual scenes, automatic execution of skilled actions) is far higher than this, but such processing does not directly support meta-audit—the essence of meta-audit is precisely the conscious interruption of automated processes and the establishment of an independent coordinate system (Chapter 3, §2.4).

In other words: this order of magnitude means that conscious information throughput runs from single digits to a few tens of bits per second—the corresponding "number of words understandable per second" depends on task and coding assumptions, and this chapter makes no fixed conversion (see §3.1 on the construct distinction).

Cowan (2001) bounded the capacity of working memory at approximately 4 chunks, consistent in order of magnitude with 10 bits/s—if one reckons a buffer of about 4 chunks, 2–3 chunk updates per second (i.e., the whole buffer refreshes roughly every 1.3–2 seconds), at 3–5 bits per chunk (**this book's own derivation; Cowan's paper gives only chunk counts and makes no bits conversion**), the two are compatible in order of magnitude (this is a derivation check, not independent validation—the chunk-update-rate assumption itself has not been independently measured). On the visual side there is a converging line from a different methodological tradition: capacity estimated by change detection varies across stimulus classes from about 1.6 objects (shaded cubes) to 4.4 (colors), with an extrapolated upper bound of approximately four or five objects as information load approaches zero (Alvarez & Cavanagh, 2004)—consistent in order of magnitude with Cowan's chunk bound, providing a check line for the denominator independent of chunk theory (the 'information load' in that paper is proxied by visual search rate, not a Shannon-bit conversion). Von Neumann (1958/2012) pointed out that the arithmetic accuracy of the human nervous system is only about 2 to 3 decimal digits.

10 bits/s is the **representative summary value** of the evolutionary architectural constraint (a task-family summary rather than a task-level upper bound; see the opening of this section; its evolutionary context is given by Sacco (2025)), not the actual operational value. Cognitive fatigue, attention fragmentation, dopamine system remodeling, and other factors **may, in theory**, cause the effective processing capacity actually available for meta-audit to be systematically lower than this order of magnitude. **This speculation awaits direct empirical testing**, and the direct tests conducted so far do not agree: on whether people who frequently use multiple media simultaneously are more easily distracted, two replication studies and a meta-analysis found the association weak, no longer significant after correction for small-study effects (Wiradhany & Nieuwenstein, 2017); reviews judge the direction negative, the effects small, and the direction of causality undetermined (Uncapher et al., 2017; Parry & le Roux, 2019). This chapter therefore does not premise scenario estimates on a decline in effective capacity; the auditable amount is uniformly reckoned at the reference magnitude—if effective capacity is indeed lower, the within-scenario ratio is only higher, and the direction of the conclusion is unchanged. This means that if the effective capacity actually available for meta-audit falls below this reference magnitude, the effective pressure facing the audit is correspondingly higher than the estimate based on the reference magnitude. This variation in availability is carried by meta-audit capacity (A) in the complete form in §4 of this chapter—covering both state-dependent, reversible variations such as fatigue and attention fragmentation and the long-term de-skilling along the automation-paradox path of Chapter 3, §4.2; the criterion for assigning a change to meta-audit capacity (A) is the meta-audit capacity callable for a given task, not reversibility. Individual differences arising from injury, disease, aging, or long-term neural remodeling do not contradict the approximate species-level constancy of the human biological substrate (E_c) on the civilizational time scale; if they enter a particular scenario analysis,

they require a separate individual-level effective-capacity variable, recorded under neither the human biological substrate (E_c) nor meta-audit capacity (A), with direction, duration, and reversibility verified separately—this chapter makes no unified claim about them. **The human biological substrate (E_c), once fixed by evolution, has its magnitude not raised by technological environments on the civilizational time scale**—no natural mechanism lifts the order of magnitude of conscious throughput as a whole; task-level estimates may exceed 10 (Table 1) but remain within the same order of magnitude. What can be eroded by the environment is the effective processing headroom and the opportunities for work that meta-audit capacity (A) can draw on, not the human biological substrate (E_c) itself; the mechanisms of this degradation have already been unfolded in Chapter 3, §4.2.

Conclusion: The denominator is stable in order of magnitude. 10 bits/s, as the representative order-of-magnitude summary given by the Zheng and Meister (2025) review, is approximately constant on the time scale of civilizational history—the pace of biological evolution (millions of years) is far slower than the pace of technological evolution (years to months). This constancy is not a design defect awaiting repair. Sacco (2025), from the perspective of evolutionary mismatch, argues that the rapid heuristics in human cognition are highly adaptive solutions in ancestral environments and only manifest as constraints in modern technological environments. The constancy of the human biological substrate is a product of evolutionary selection—this means the rise of the cognitive Reynolds number is a unilateral race: the numerator, the media conductivity coefficient (Λ), accelerates on a yearly scale; the denominator, the human biological substrate, remains stationary on a million-year scale.

This book uses the human biological substrate (E_c) not as a constant to be precisely calibrated but as a directional constraint: it is extremely small, and it is constant. The robustness of this directional judgment rests on two facts: first, conscious processing is far smaller than sensory input (by roughly eight orders of magnitude), showing that the human biological substrate (E_c) is robustly a bottleneck—this fact serves only to establish the bottleneck's standing and does not enter the ratio (the numerator of the ratio is audit-awaiting arrival, not the raw sensory stream, §2.2); second, the human biological substrate (E_c) is not raised by technology on the civilizational time scale, while audit-awaiting arrival bears rising pressure along the media trajectory examined here. The direction is decided by the latter contrast; regardless of how the specific value is revised, this qualitative conclusion remains unchanged. To concretize the boundary of this robustness: even if the human biological substrate (E_c) is taken at the upper end of that review's Table 1, about 50 bits/s (see the opening of this section), the gap between human conscious processing and sensory input would still be over seven orders of magnitude—the directional judgment of this chapter is robust to values of the human biological substrate (E_c) across the range listed in that table.

§2.2 The Media Conductivity Coefficient (Λ): Irreducible to a Single Scalar

Chapter 3, §2.2 defines the media conductivity coefficient (Λ) as the efficiency with which information reaches the host, and distinguishes three layers: macro infrastructure, meso distribution architecture, and micro contagion dynamics. This multi-layered structure means the media conductivity coefficient cannot be reduced to a single bits/s value. The same information arrival rate of 100 bits/s, delivered through algorithmic personalized targeting versus undifferentiated public broadcast coverage, may have markedly different consequences for attention occupation and subsequent judgment. The media conductivity coefficient is not only channel capacity but also targeting precision—the capacity of information to penetrate the host's specific cognitive vulnerabili-

ties.

Although irreducible to a scalar, a set of proxy indicators can be constructed for different historical stages to anchor, in order of magnitude, the trends of the media conductivity coefficient:

- **Audit-Awaiting Arrival**: The readings awaiting calibration that arrive at the host per unit time—propositions, numbers, citations, instructions—reckoned as the time required to audit them. The reading criterion is treated as an ideal type: whatever arrives counts as a reading, without filtering by "whether it is accepted as true or acted upon"—that judgment cannot be effectively assessed at the moment of arrival (acceptance may occur now, at the moment of purchase, much later, or never). The Fermi estimation of this chapter counts this dimension in audit labor time: the ratio of the text quantity arriving per hour to ordinary reading per hour, the split into the two components, and the scenario estimates, see §2.3.

Pavlov et al. (2008), in a system-dynamics model, defined it as the degree of overload: no overload when the time required does not exceed the time available, overload when it does (Pavlov's model uses the time actually available for the task; this chapter takes a one-hour measurement window—the ratio depends only on arrival rate and reading speed and is independent of window length—the two differ when the host actually allocates less than the full window). The human biological substrate (E_c) does not appear here as a divisor; it acts in two places: the time to locate and retrieve evidence can shorten with better tools and upstream calibration (the source coefficient of §3.3 covers this layer), whereas the time to read the evidence and make the comparison does not shorten with technology; verification work actually completed within the window cannot exceed elapsed time. When the same host compares two media, the reading speed is the same, and the comparison reduces to the ratio of the two arrival quantities, read in orders of magnitude (§1.2). The raw quantity of information at the sensory layer is far greater than this, but it does not constitute an audit load, and this chapter does not use it to compute the ratio. This is a proxy for the rate dimension. - **Control over Reception** (compulsoriness of information arrival—the lower the control, the higher the compulsoriness): Whether the host can control the rate, sequence, timing, and release timing of information arrival. The migration of control from the host to the projecting party means a change in the opportunity structure by which information enters sustained attention and subsequent judgment. Monge Roffarello et al. (2023) classified attention-capture deceptive designs in digital interfaces into eleven types, among them infinite scrolling, autoplay, and recapture notifications; for the autoplay type the common point is that no pause cue is ever given—and the shared effect of such interface designs is to erode the host's natural "stop" signal (Ye, 2025). A meta-analysis of media multitasking found the cognitive costs larger the less control users had over the pace and sequence of the information flow (Jeong & Hwang, 2016). - **Verification Window**: The time available to the host after receiving information for verification, reflection, and cross-source comparison. The compression of the verification window is equivalent to meta-audit capacity operating at reduced frequency in a high media conductivity coefficient environment—for short videos, what is compressed is absolute time (mostly a few seconds to 60 seconds per video, Jain et al., 2025); for AI generation, absolute time is ample. - **Trigger for Checking**: Whether the host initiates verification, reflection, and cross-source comparison after receiving information. For short videos, continuous context switching may inhibit initiation; for generative AI, semantic coherence and default acceptance may reduce the frequency of initiation—a mechanism hypothesis whose empirical basis is still accumulating (§4). The trigger records whether checking is initiated, checking work records whether back-and-forth comparison actually occurs, and meta-audit capacity (A) records the capacity available for a specified task; the three are related but not identical. When checking is not initiated, the corresponding work does not occur, but this establishes neither a loss of capacity nor a

general tendency toward zero. Friction here takes the necessary-condition sense of the cognitive friction of Chapter 3: the work of back-and-forth checking and adversarial correction between host and information—no work, no audit—distinct from the interface-resistance sense of "adding or reducing friction" in the digital nudges. The trigger is a subjective disposition to initiate, while the verification window is objective available time. - **Personalization Precision**: The targeting precision of the projecting party toward the host's individual cognitive vulnerabilities. Fagan's (2024) review cites a seven-type classification of digital nudges—information provision; framing; salience, reminders, and prompts; defaults and commitments; adding or reducing friction; reinforcing behaviors with feedback, rewards, and punishments; and engaging users emotionally—of which defaults and commitments, friction adjustment, and behavioral reinforcement directly fine-tune the penetration efficiency of the media conductivity coefficient on the individual host.

Eye-movement measures are themselves proxy evidence for reading processes. Fan and Liu (2007) caution that eye movement is not a unique or perfect reflection of reading psychology and should be combined with traditional behavioral experiments, event-related potentials, and functional magnetic resonance imaging. Accordingly, this chapter uses reading rate, rereading, and skipping to estimate how arrival is processed, not to determine meta-audit capacity directly; the latter still requires records of external verification and cross-source comparison.

Comparison across the five dimensions (the first row gives the direction and basis of each scenario's ratio, the remaining rows are ordinal judgments; **none is a precise measurement**):

Table 1. The Five Proxy Dimensions of the Media Conductivity Coefficient and Comparison under Default Usage Modes

Proxy Dimension	Books	Television	Short Videos	Generative AI
Audit-Awaiting Arrival (reckoned in the time auditing requires, compared with the length of the arrival stream itself)	Reader sets the pace; can be pressed below the auditable amount	Broadcast sets the pace, no looking back; arrival not audited in time is lost as it passes	Single items last from a few seconds to 60 seconds; under the protocol of continuous playback filling a one-hour window, the ratio exceeds 1 when the verification component, converted at its restoration references, exceeds the margin left by reading	Generation volume grows with speed and the number of parallel sessions; audit time accumulates per item; the ratio rises with generation volume

Proxy Dimension	Books	Television	Short Videos	Generative AI
Control over Reception (whether the start, pause, sequence, speed, and release timing of arrival can be decided)	High	Medium	Low	Low
Trigger for Checking (whether checking is initiated)	Depends on source, task, and reading practice	Depends on source and viewing context	Continuous context switching may inhibit initiation	Default acceptance may inhibit initiation
Verification Window	Not constrained by the medium's mechanics	Minutes	Seconds	No per-item time limit; total audit time falls short as generation volume grows
Personalization Precision	No real-time individual targeting	No real-time individual targeting	Extremely High	Extremely High

Table note: the trigger-for-checking row states mechanism hypotheses, not measured grades; none of the cells presupposes a general tendency toward zero initiation.

The first row of Table 1 gives only direction and basis; the values are computed within specific scenarios, and their reading is constrained by the short-time-slice boundary of Chapter 3, §5.2: numerator and denominator come from the same scenario and the same unit. The web is an example with published numbers. Privacy policies on popular websites average about 2,400 words (Steinfeld, 2016); at silent-reading speed one policy takes about 10 minutes to read (this book's derivation), consistent with the estimate of about 10 minutes per policy in McDonald and Cranor (2008); on that basis they estimated that reading word-for-word the privacy policies of every website visited would take each person about 244 hours a year—about 40 minutes a day, slightly more than half of the roughly 72 minutes a day then estimated for Internet use—the reading workload of this one class of terms alone amounts to more than half of daily Internet time. The 72 minutes are the arrival window, not the host's budget for verification; the comparison shows how large the reading workload of such terms is, not that audit capacity is thereby filled. The audit time a single item requires is also under the projecting party's control: firms that do not respect privacy deliberately write their policies obtusely to make testing a burden; the more such firms there are, the higher the cost of testing and the less willing consumers are to test (Vila et al., 2004). How default settings change actual reading investment is described in the web experiment of §3.2; actual investment is not used to back out audit capacity. Subtitles are another example (§3.2): their arrival speed and silent-reading speed can be compared in the same unit. Ratios from different scenarios are compared only in their positions relative to their respective capacity boundaries, not on a single unified axis.

Arrival at the sensory layer far exceeds conscious processing (§2.1), but it is not an audit load; video bitrate describes the encoded transport stream and is likewise not audit-awaiting information. This chapter therefore gives no multiples of the human biological substrate (E_c) for audiovisual media. Each medium is still reckoned in its **default usage mode**—books at a pace set by the reader, audiovisual media at a pace set by playback—and **who sets the rhythm is itself a manifestation of the difference in control** (§3.1).

The audit-awaiting arrival that exceeds the auditable amount is not "audited" but bypassed—it either enters subsequent judgment without verification or passes by in an unverified state; a portion of it may operate through the default mode network (DMN) and the reward system and other unconscious processing pathways (the functional magnetic resonance imaging (fMRI) evidence of Su et al. in §3.3 is compatible with this picture). The turbulent interval is not determined by a single rate threshold: beyond the arrival-rate dimension, control over reception, the trigger for checking, and personalization precision jointly determine interval membership. Take television as an example: its arrival is paced by broadcast and cannot be replayed—once a piece of information has left the screen or ceased to be heard, it is no longer available to be encoded (Lang, 2009)—but its trigger for checking is not uniformly downgraded on this basis and it lacks personalization calibrated in real time on individual behavioral data, so it falls in the transition interval; short videos and generative AI enter turbulence through absolute-time compression and the possible strong suppression of the trigger for checking (an inference), respectively (the two-type distinction in §6.3). This multi-dimensional determination is consistent with the ordinal comparison of §1.2—each dimension is judged separately, not on a single unified axis.

The five dimensions may interact—the combined effect of low control over reception, high personalization precision, and a short verification window may not be a simple sum of the individual effects (this interaction structure is unverified; the chapter makes no quantitative aggregation on this basis). This is the primary reason the media conductivity coefficient cannot be reduced to a single scalar and also the reason Fermi estimation can only give orders of magnitude, not precise values.

The engineering shaping of the above five dimensions has a comparable mature precedent in the video game industry. Swink (2009) delimits "game feel" as the tactile, kinesthetic sense of the player manipulating a virtual object—the sensation of control in a game—and gives measurable conditions for real-time control, with the human-cycle figures cited from the three-stage cycle of Card, Moran, and Newell's (1983) Model Human Processor: **Perceptual Processor approximately 100 ms (50–200 ms), Cognitive Processor approximately 70 ms (25–170 ms), Motor Processor approximately 70 ms (30–100 ms)**, together constituting a complete correction cycle of approximately 240 ms. Swink's secondary presentation swaps the latter two ranges; the values here follow Card et al.'s original text. From this, Swink divided the response latency of the system to input into three threshold grades: **at 50 ms the response feels "instantaneous"; above 100 ms the lag is "noticeable but ignorable"; at 200 ms the response feels sluggish**; meanwhile, the cycle of continuous interaction is required to remain stable at 100 ms or below, otherwise "the flow of control is broken" (Swink, 2009).

These thresholds are not the aesthetic preferences of game designers but the engineering projection of the human perception–cognition–motor closed loop—the game industry, in the 2000s, already wrote them into engineering manuals with millisecond-level precision. This same set of refined perception–feedback engineering was later transplanted into digital persuasion and short-video products as an engineered means of "making it impossible for the host to stop"—response latency is compressed below the perceptual threshold, so that the next interaction begins before awareness—whether this design lowers the probability of stopping or re-checking requires separate behavioral testing. This is the material basis, at the product engineering layer, of what §2.2 calls "the

media conductivity coefficient cannot be reduced to a single scalar": the multi-dimensional shaping of the media conductivity coefficient is not an abstract concept but an engineering practice for which millisecond-precision reference books already exist.

§2.3 Fermi Estimation: Three Progressive Stages and Scenario Estimates

Ideal Type and Textual Conversion

The cognitive Reynolds number compares the information-arrival pressure of a media scenario with the host's processing capacity (Chapter 3, §5.2). The Fermi estimation of this chapter assesses that pressure under ideal-type assumptions: convert the content arriving in each media scenario into text, convert it into an audit burden at the host's ordinary reading speed, and compare it with the length of the arrival stream itself. Three rules of the ideal type: whatever arrives counts as a reading, without filtering by "whether it is accepted as true or acted upon" (§2.2); the reading speed is ordinary silent reading—238 words per minute in English, 14,280 words per hour (Brysbaert, 2019; the Chinese baseline is 320 characters per minute, a median scenario, see §4); the measurement window is uniformly one hour, and the ratio depends only on arrival rate and reading speed, not on window length.

The conversion for each medium: for books and long texts the arrival is itself text, counted directly in words; for speech media the arrival is filled by the playback rate—normal speech is 150–200 words per minute (Eysenck & Brysbaert, 2018), so one hour of broadcast transcribes to 9,000–12,000 words; factual assertions in the visual track (numbers, persons' identities, product functions, action instructions) are converted into word equivalents, and the same meaning expressed simultaneously by the audio track, subtitles, and images is counted once; the visible output of generative AI is counted as characters converted from tokens (the average characters per token is the conversion coefficient).

The First Formula

The text quantity arriving within one hour that requires processing by the host is the arrival equivalent (W_s), and the host's ordinary reading in one hour is R_s. The first formula:

$$\lambda_s = \frac{W_s}{R_s}$$

λ_s is the ratio of the labor time required to process one hour of arrivals to that one-hour arrival stream itself; a ratio greater than 1 means the audit burden of the arrivals exceeds the length of the arrival stream itself, and 1 is the capacity boundary of that scenario (Chapter 3, §5.2).

Two Channels and Compression–Restoration

The arrival equivalent splits into two components, each carried by one channel, both reckoned as text and each divided by the reading speed. **Channel A (arrival→reading)** carries the reading component: whatever surface text arriving must be read once—the audio track transcribed at speech rate, generated text counted in characters—enters the reading equivalent ($W_{A,s}$). The reading component merges only multimodal repetition at the same moment: the voice speaking, the subtitles showing, the images demonstrating are naturally merged into one pass in comprehension; repetition across time is counted in full—every viewing occupies reading labor, and

repetition itself raises acceptance and suppresses the initiation of checking (§3.3). Source credibility changes nothing in this component—a thousand untrustworthy characters are still a thousand characters.

Channel B (arrival→verification) carries the verification component, reckoned by **compression–restoration**.

> Laws, like sausages, cease to inspire respect in proportion as we know how they are made. (Saxe's new lecture, as reported by the McKean Miner, April 22, 1869; as cited in Shapiro, 2006—commonly misattributed to Bismarck; the Yale book of quotations notes that "Saxe" may refer to the lawyer-poet John Godfrey Saxe)
>
> The variant circulating in IT standards circles (Gutmann): People who love sausages, respect the law, and work with IT standards shouldn't watch any of them being made.

The host's situation differs from both sentences: the person who does not watch the making receives the finished product—and the other half of the sausage metaphor is that trust without watching the making can only be delegated or held pending. The audit-awaiting content of arrivals is precisely the finished product of layered processing: a one-minute account of a two-hour film, a paragraph summarizing a paper, one list aggregating a set of component data. The accounting method of compression–restoration: the verification equivalent of one compressed proposition is reckoned by the **restoration reference** required to check it—to check "that video's rendering of the film" to the point where truth or falsehood can be judged, one needs a reference that accounts for the film (sufficient plot and commentary text); to check that summary of the paper, one must return to the original paper. Restoration does not require the host to audit—the host may decline to audit; what the formula registers is the price of compression: the shorter the consumption time is compressed, the greater the restoration responsibility in proportion. Watching the two-hour film, the two hours are booked in the reading component (viewing is processing) and the verification component is near zero; scrolling the one-minute account of that film, a one-hour window fits roughly sixty items, each carrying a restoration reference—the verification component is converted from the text quantity of the restoration references. The same quantity of information, compression moves it from the cheap reading component (processed at reading speed) into the expensive verification component (reckoned by restoration reference)—this is the formula form of the risk brought by compressed time, and the reason the two components are booked separately and not merged into a single multiplication: books are reading-dominated, commentary-style videos verification-dominated, and merging into one multiplication would erase the structural difference between the two classes of media.

The Verification Component Reckoned by Propositions

The unit of account of the verification component is the **proposition**. Extract checkable propositions from the arriving content, deduplicate at the proposition level—the same proposition is checked only once, covering its subsequent repetitions; shared evidence and reusable verification results are counted once—and reckon the restoration reference per item: the verification equivalent ($W_{B,s}$) equals the verification item count (N_s) times the restoration reference word count per proposition.

The word count of the restoration reference is set by backward derivation. The forward route fails: no consultable distribution exists for what reference a given video points to or how long that reference is. The backward route starts from the proposition end: **how many words does it take to state one proposition adequately?** A proposition of the natural sciences takes about 500 words—definitions, statements, and evidence can be written as a compact block of formulas and data, the domain of mathematical formulas; a proposition of the social

sciences takes about 2,500 words—chains of argument, context, and boundary conditions resist compression, and stating one causal proposition adequately requires the length of a complete argument. A 45-second commentary video transcribes to roughly 200 words (at speech rate) and carries one proposition—its restoration reference is thus about 2.5 times (natural sciences) to roughly 12 times (social sciences) the transcribed word count. Elimination checks both ends: a multiple below 1 means the arrival is longer than the reference, which is not a compressed product; multiples above a hundred belong to headline-level arrival (a dozen characters compressing one proposition), not the form of a 45-second narrative video. The back-derived restoration labor per proposition (2,500 words at 320 characters per minute) is about 8 minutes, consistent with the intuitive range of "several minutes per item." Where restoration must go to the source of the reference itself—re-watching the original film, re-checking the original data—the labor takes the length of the source, and the back-derived word count is its lower bound.

How many propositions each video carries is set by its persuasive structure. A short video aimed at behavioral conversion is constructed on three elements—trigger (the hook), motivation (the content), and behavior specification (the call to action):

> behavior is a product of three factors: motivation, ability, and triggers… for a person to perform a target behavior, he or she must (1) be sufficiently motivated, (2) have the ability to perform the behavior, and (3) be triggered to perform the behavior. (Fogg, 2009)

The hook's suspense assertion, the content's informational assertion, and the call to action's commitment assertion are each checkable propositions—the baseline scenario takes three per video (an explicit assumption; for purely recreational streams—music, dance, stunts—the checkable propositions approach zero, and their high ratios stem mainly from the reckoning convention rather than the content, see the table note in the next section). This structure shares its origin with the expressive shaping of the platform age: hook openings, emotional endings, and rhetorical fragments replacing chains of argument (Chapter 27, §7).

The restoration reference of a proposition is discounted by its **responsible party**. Whether anyone is answerable for a proposition determines how much of the verification burden upstream calibration has covered: a thirty-page newspaper has an editorial board and bylined reporters—the pre-publication editorial process is calibration work already performed; a book has an author—the author is answerable for the content through signature and professional reputation. For propositions with a responsible party, the host's residual verification burden is discounted (low coefficient); for propositions without one—anonymous accounts, unsigned generated content—upstream calibration is absent, and the host must restore to the original reference in full (high coefficient). The coefficient is binary: propositions with a responsible party take the baseline; those without take three times and up—the work performed by the responsible party (author self-checking plus the editorial process) conservatively covers more than half of the calibration workload, to which the prior of distortion for anonymous sources is added; no upper bound is set, and the weakest extreme of the responsibility structure (a missing recourse channel, i.e., the missing compensation loop) is discussed in other chapters of this book. The discount of the responsible party holds by domain: a responsible party who signs within their own field enjoys the discount; speaking outside their own field—an expert in field A discussing field B—the discount lapses and the coefficient turns high, credibility ascription having always been confined to the subject taught (§3.3, Rieh & Hilligoss, 2008). The responsible party acts on the verification component only: source credibility changes nothing in the reading component, and a thousand untrustworthy characters are still a thousand characters. Generative AI is reckoned as having no responsible party (coefficient taken high), and the attribution

of responsibility for generation systems is left to later chapters. The same proposition with the same reference word count, arriving via a bylined newspaper versus via an anonymous video, differs by at least a factor of three in verification equivalent—the difference lies not in the content but in whether anyone shares the calibration.

Assertions and advertisements are not distinguished: commitment-type content—claims about future delivery, advertising, propaganda, placement, and optimization content relayed through generative engines—is uniformly treated as a subtype of proposition, with no special parameters; advertising character changes nothing in how the proposition is booked. Whether commitment-type propositions constitute cognitive debt is recorded separately by the §7 protocol (accepted, retained, and entering subsequent invocation before entering the debt ledger), not decided at the estimation layer. The first formula accordingly expands:

$$\lambda_s = \frac{W_{A,s}}{R_s} + \frac{W_{B,s}}{R_s}$$

Structural Propositions

Four structural judgments follow directly from the above expression, plus one criterion.

For media whose reader sets the pace, the reading component is structurally non-overloaded. The arrival of books and newspapers is filled by the reader's own reading speed: $W_{A,s} \leq R_s$—one cannot read more than one reads, and the reading term does not exceed 1. This is the arithmetic form of "books and newspapers are not unlimited"—the medium does not automatically generate arrivals exceeding the reading speed, with an hourly cap of 14,280 words.

For speech media whose broadcast sets the pace, the reading term is always below 1. The arrival of radio, television, and short-video audio tracks is filled by the playback rate (r, 150–200 words per minute): $W_{A,s} = 60r$, and the reading term $60r/R_s$ lies between 0.63 and 0.84. Listening is slower than reading—an hour of broadcast speech, once transcribed, takes less than an hour to read. Comprehension is not the bottleneck.

For generative media whose machine sets the rate, the reading term is not capped by consumption. The reading equivalent of generative AI ($W_{A,s} = G\alpha$, tokens within the one-hour window times the character-conversion coefficient) grows with generation speed and the number of parallel sessions, unconstrained by human consumption speed—the generation volume takes the range of 5,000–200,000 tokens per hour (light sessions to heavy parallelism; the demonstration case $G = 100{,}000$, $\alpha = 4$, with reading term 5.6 at the midpoint—§4).

The overload criterion is a counting criterion. $\lambda_s > 1$ if and only if the verification component exceeds the margin left by reading: $W_{B,s} > R_s - W_{A,s}$. For the first two classes of media the margin lies between 0 (books, reading exactly filling the window) and 0.16–0.37 times the reading speed (speech media); the verification component has a structural numerical source—item count times the restoration reference per item, and 45-second commentary videos at tens of items per hour, each carrying three propositions and 500–2,500 words of restoration reference per proposition, push the verification component tens of times beyond the margin.

Overload is a necessary, not a sufficient, condition for turbulence—the second criterion of deferability. Under the ideal-type criterion (whatever arrives counts as a reading), the full verification demand of any medium's hour exceeds one hour, books included (§3.1: the laminarity of books lies not in the calibration demand of one hour of normal reading necessarily falling below one hour, but in the reader's dual control over

arrival and checking). What distinguishes laminar from turbulent flow is the second criterion: can unchecked readings be deferred. The text of a book stays in place and the verification window does not close, so the excess becomes a stock that can be digested in installments (the transition-type debt trajectory of §7); a short video closes as it scrolls away and the per-item verification window is measured in seconds, so the excess sediments directly (the turbulent-type trajectory of §7). Turbulence equals overload plus infeasible deferral; λ_s alone does not determine interval membership.

Estimates for the Principal Media Scenarios

The quantities of the scenario estimates are set in two classes: what can be anchored is anchored—reading speed, speech rate, usage statistics, per-item duration (all with literature or datasets); what lacks an anchor uses explicitly stated Fermi assumptions—restoration reference word count and proposition count, and the sensitivity check consists of varying these assumptions; no constructive protocol is adopted. The measurement window is uniformly one hour (the ratio is independent of window length; the same window is taken so the scenarios can be displayed side by side). The reading speed is 14,280 words per hour in English; Chinese scenarios use the 19,200 characters-per-hour baseline (320 characters per minute, median scenario), see §4.

Table 2. Fermi Estimates within a One-Hour Window (Literature-Anchored Ranges; Parameterized Results, Not Measured Platform Averages)

Scenario	Anchoring of Arrival	Reading Equivalent ($W_{A,s}$)/hour	Reading Term	Verification Item Count (N_s)/hour	Deferability
Books	Reader sets the pace (Brysbaert, 2019)	14,280 words	1.00 (by definition)	open	high (text persists)
Newspapers	Same class as books: reader sets the pace	≤14,280 words	≤1.00	open	high
Radio/television	Broadcast sets the pace (Eysenck & Brysbaert, 2018)	9,000–12,000 words	0.63–0.84	open	low–medium (no replay, Lang, 2009)
Short videos (audio-track criterion)	Usage statistics (see text)	9,000–12,000 words	0.63–0.84	45–240 items	extremely low (closes as it scrolls away)
Generative AI	Generation-volume range (see text)	20,000–800,000 characters	0.28–11.2	open	medium (working records persist)

Table note: the reading term equals the reading equivalent divided by ordinary reading per hour; reading labor is that value times 60 minutes (0.63–0.84 corresponds to 37.8–50.4 minutes). The word equivalent of visual assertions does not adopt a constructive protocol; its bottom-up anchoring procedure is left for later. The verification

component is demonstrated by explicit scenarios: the item count takes the anchored range of 45–240 per hour, three propositions per item (Fogg's three elements), and 2,500 words of restoration reference per proposition in the social sciences—demonstration combinations: 80 items×3 propositions = 240 propositions per hour, reference 600,000 characters, verification component about 31 (responsible-party baseline); anonymous arrival at the three-times responsible-party coefficient about 94; knowledge-commentary type (natural-science reference 500 words, signed author) about 6–7 (explicit scenarios, not measurements; for purely recreational streams the propositions approach zero and their ratios stem mainly from the reckoning convention). The AI row takes the range of 5,000–200,000 tokens per hour (light to heavy parallelism, with the demonstration case of 100,000 tokens at the midpoint—§4) and a character-conversion coefficient of 4. The finer parameters of compression–restoration (source credibility, type intervals of the visual equivalent) are not unfolded in this chapter. The 1 in the books row is constructed by continuous reading, not an empirical laminar threshold. The rows do not form a unified cross-media scale. The acceptance, retention, and subsequent invocation of unchecked commitments are recorded separately in the §7 debt ledger.

Three anchor groups for short videos. Usage duration: market statistics show TikTok average daily use rising from 27.4 minutes in 2019 to 55.8 minutes in 2023 (as cited in Jain et al., 2025); adolescent samples run higher and are cross-nationally consistent—Greece 2.77 hours/day, the United States 2.17–2.5 hours/day, China 2.85 hours/day (Bilali et al., 2025 and its citations). The two samples differ in population (market statistics of all users versus adolescent self-report) and are not averaged together; this table takes "one active hour of use" as the window scenario, with daily figures serving only as scale reference. Per-item duration: a study of 112,610 videos on algorithmic amplification gives a mean duration of 79 seconds and a viewing completion rate of about 64% (Baumann et al., 2025); the review convention is that single videos mostly run from a few seconds to within 60 seconds (Jain et al., 2025). Items per hour: reckoned as continuous viewing over one active hour, $N_s = 60/(\bar{d}f)$—mean duration ($\bar{d}$) in minutes and completion rate (f) as the proportion actually watched—the anchor combinations give 45–240 items per hour, with a midpoint of about 140; the interval midpoint, not a measured platform average.

A demonstration of compression–restoration (Chinese-reckoned): one hour of continuous 45-second commentary videos, about 80 items, each transcribing roughly 200 characters and carrying three propositions—reading term about 0.83 (audio track 16,000 characters against 19,200); verification component by the back-derived restoration reference: knowledge-commentary type (natural-science reference 500 characters, signed author) about 6; sales-and-traffic type (social-science reference 2,500 characters, anonymous arrival, responsible-party threefold) about 94—λ spans from about 7 to about 95, an order of magnitude across content types and responsibility structures. Watching the two-hour film in the same window instead (two windows): reading term 1, verification component near zero. The information of the same film, arriving as the original or as commentary, differs in audit burden by one to two orders of magnitude—the difference lies not in the total quantity of information but in which component compression has moved it into, and who signs to share the calibration.

Reuse of estimates already cited in the chapter. Privacy policies—policies on popular websites average about 2,400 words (Steinfeld, 2016), each taking about 10 minutes to read through (McDonald & Cranor, 2008), and reading word-for-word the policies of every website visited would take about 244 hours a year, about 40 minutes a day, against roughly 72 minutes of daily Internet use (as cited above)—the reading workload of this single class of terms alone amounts to more than half of daily Internet time; an instance of "engineered arrival" in the reading component: the arrival quantity is designed, not filled naturally by consumption. Subtitle speed (raised

from 12 to 17–20 characters per second, Szarkowska & Gerber-Morón, 2018) is a same-structure verification of broadcast-paced media: subtitle arrival speed is calibrated to silent-reading speed, reading term ≈ 1.

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§3 The Three Intervals of the Cognitive Reynolds Number (Re_c): Directional Judgment, Not Precise Measurement

What follows does not give precise numerical values of the cognitive Reynolds number (Re_c) for each media stage—that would violate the epistemological constraint declared in §1.2. Instead, the qualitative direction of the transition from laminar flow to turbulence is anchored by three intervals. The basis for the judgment of each interval is the most reliable type of evidence within that interval, not a uniform calculation method; cross-interval comparison is therefore directional and dimension-by-dimension, not a continuous ordering on a single scalar axis.

§3.1 The Laminar Interval: The Host Can Regulate the Rhythm of Comprehension while Auditing Conditions Begin to Diverge

A conceptual boundary must first be declared: the laminar/turbulent judgment of this chapter targets the **default usage mode** of the medium—i.e., the mode of use that the engineering of the medium guides—rather than an inherent property of the medium itself. A copy of the *Critique of Pure Reason* on a shelf has enormous information density, but the act of reading it reduces it to the laminar range. The default usage mode of books keeps control in the hands of the host; the default usage mode of short videos migrates control to the algorithm. This distinction is crucial for the argument of "migration of control" in §5. A further distinction between two processing rates is needed: the comprehension rate (understanding, reading what has arrived) and the audit rate (verifying against external references, cross-checking sources, building an independent coordinate system). The human biological substrate (E_c, approximately 10 bits/s) is the common order-of-magnitude reference for both (task-level estimates may exceed it, Table 1), but auditing occupies more conscious operations per unit of input than comprehension, so the same arrival stream may be laminar for comprehension and turbulent for audit. Judgments anchored below on comprehension speed assert only the comprehension-rate ratio and control over arrival; estimation of the audit-rate ratio must proceed by the calibration time of §2.2–§2.3—for example, by the more-than-twofold conversion from study-style reading below 100 words per minute (Eysenck & Brysbaert, 2018) to ordinary reading at about 238 words per minute (Brysbaert, 2019) (this book's derivation)—the two do not substitute for each other.

Books. The information arrival rate of reading and the human biological substrate are on the same order of magnitude. Brysbaert's (2019) meta-analysis gives the average silent reading speed of adults as approximately 238 words per minute (wpm) for English non-fiction texts. Shannon (1951) estimated the entropy of English under long-range statistical constraints at about 1 bit per letter, with roughly 75% redundancy; Cover & King (1978), using a gambling-based estimator, give an independent estimate of approximately 1.3 bits/symbol, in agreement with Shannon's order of magnitude. But the information rate of character encoding and the semantic processing of reading comprehension are different constructs; this chapter makes no numerical conversion between them. The modulation of effective processing rates by semantic structure has a direct experimental instance: under rapid serial presentation, viewers can recall entire sentences presented at 12 words per second, whereas only two or three words can be recalled from unrelated word lists presented at the same rate (Potter,

2012)—at a constant arrival rate, how much can be processed depends on semantic structure; this is experimental evidence of the same family as the construct distinction above. The key lies not in the precise bits/s value but in a structural fact: the rhythm of arrival is under the reader's control. Readers can look back—when processing runs into difficulty they can reread the text just read, whereas an auditory signal is lost once it has decayed from echoic memory; when verification is needed, readers can further slow to study-style reading, below 100 words per minute (Eysenck & Brysbaert, 2018), while the silent-reading speed for ordinary reading is about 238 words per minute (Brysbaert, 2019)—item-by-item verification therefore takes more than twice the time of ordinary reading (this book's derivation). (Reading-rate conventions: the ordinary-reading component of the scenario estimates uses silent prose reading at about 238 words per minute, or about 20 English characters per second. This figure includes pauses and rereading and represents an ordinary-reading pace, not the total auditing pace after external verification is included. The average reading rate inferred from fixation durations is about 28–45 English characters per second (Zheng and Meister, 2025); it illustrates a different measurement method, is not a momentary processing ceiling, and enters neither Table 2 nor the baseline ratio in §7. Ordinary-reading time in §7 uses the task rates of §2.3, with verification time recorded separately. Character counts use a consistent convention for spaces and punctuation across subtitle rates, characters per word, and characters per token; Chinese characters, segmented Chinese words, and English words are not treated as interchangeable units. The two rates must not be substituted for each other.) The laminar character of books lies not in the calibration demand of an hour of ordinary reading being necessarily below one hour—under the ideal-type criterion (whatever arrives counts as a reading, §2.2), the item-by-item calibration demand of an hour of ordinary reading usually exceeds one hour—but in the reader's dual control over arrival and verification: stopping at any time presses down the inflow of new arrivals; allocating calibration time at the reader's own pace governs the digestion of the stock; and deferring verification—it does not reduce the calibration demand of readings already arrived in the current period (demand = the time required to audit all readings arriving within the period, independent of when verification occurs), with the deferred portion carried into later periods as pending obligations (§7)—yet deferral is possible only because the text stays in place and the verification window does not close, which is itself part of the control structure. Laminarity rests on who controls arrival; this is the basis of the §3.2 criterion (migration of control).

Telephone and Radio. Coupé et al.'s (2019) systematic comparison of 17 languages revealed a constant: despite enormous differences in syllabic rates across languages, the **speech-encoding information** transmission rate always converges to approximately 39 bits/s. A distinction between two different constructs is needed: 39 bits/s is the **speech-encoding information rate of the channel** (the output rate at the transmitting end), whereas the approximately 10 bits/s used as the denominator reference in §2.1 is an order-of-magnitude summary of behavioral information throughput (Zheng & Meister, 2025; not a measured audit-task rate); the two are not the same quantity. This constancy is itself a product of the human biological substrate—language evolution has already matched the output rate to the listener's processing bottleneck. (This interpretation goes beyond the scope of the original conclusions of Coupé et al.—the original work emphasizes the constancy of information rate across languages itself, rather than attributing it to a single listener processing bottleneck; the Uniform Information Density (UID) hypothesis of Jaeger (2010) likewise starts from a channel of limited capacity, examining how speakers avoid local information peaks, without reducing the cross-linguistic constancy to a single listener bottleneck—a more general alternative explanation. This chapter adopts the processing bottleneck interpretation to connect to the denominator argument of the cognitive Reynolds number.) The telephone introduced synchronous interaction—both parties in a conversation must respond in real time, unable to pause as in reading,

though they can ask the other to repeat or slow down. Everyday speech runs at about 150–200 words per minute (Eysenck & Brysbaert, 2018), the same order as silent reading. Linear broadcasting affords no asking for repetition: once information ceases to sound, it is no longer available to be encoded (Lang, 2009), and arrival not audited in time is lost as it passes. As regards the comprehension rate, broadcasting remains laminar—its speech rate is of the same order as everyday speech; but it can be neither paused nor replayed, and content requiring verification loses its online verification window the moment it airs—a trait already characteristic of the transition zone (§3.2). This chapter places it at the upper boundary of the laminar range in the sense of comprehension rate and arrival rhythm, not as a claim that its auditing conditions match those of pausable reading.

The common feature of the laminar interval: the rhythm of reception is not forcibly accelerated by the mechanics of the medium—reading can be entirely self-paced; the telephone permits negotiated pacing but requires synchronous responses; linear broadcasting without playback controls proceeds at the broadcaster's schedule. The three do not constitute the same control condition over reception: reading can be slowed, reread, and stopped at any time, pressing the audit-awaiting arrival below the host's own verification capacity; the telephone allows asking for repetition and negotiated pacing but requires synchronous responses; linear broadcasting's speech rate is of the same order as everyday speech—which supports only laminarity at the comprehension level, its non-pausable, non-replayable character already bearing transition-zone traits (end of this section)—meta-audit is not structurally squeezed out in reading and telephony, while in broadcasting it is subject to transitional constraints. Within this interval, the ratio and auditing conditions must be stated separately. In reading and telephony, the host can press the within-scenario ratio below 1 and meta-audit can operate normally. Linear broadcasting is laminar only at the level of comprehension; its auditing conditions already bear transition-zone features and cannot share the capacity criterion applied to reading and telephony.

§3.2 The Transition Interval: Control of the Media Conductivity Coefficient (Λ) Begins to Migrate

Film is the first industrialized audiovisual projection medium, which for the first time extended the media conductivity coefficient from the textual/auditory channel to the dual audiovisual channel. But the physical constraints of the cinema themselves constitute a built-in meta-audit window—a fixed duration of approximately 2 hours per screening, the decision cost of purchasing a ticket and traveling, and the spontaneous discussion that naturally occurs after the screening. Hanich (2018) systematically demonstrated the "audience effect" of collective cinema viewing: viewers co-present in the same physical space perceive and regulate one another—the synchronized oscillations of laughter, fear, and attention constitute a shared pre-reflective experiential background—one that provides social conditions for post-hoc discussion and reflection, though collective experience itself is not independent reflection—viewers are not only watching the film but also perceiving how others are watching the film. The social conditions this collectivity provides were stripped away piece by piece in subsequent personalized media.

Television removed the physical constraints of the cinema—it entered the home, with channels operating 24 hours a day; the decision cost of turning it off was very low, but inertia was very high. Cutting et al.'s (2011) statistical analysis of 160 Hollywood films from 1935–2010 documented this long-term trend: the Average Shot Length (ASL) continuously declined from approximately 10 seconds in the 1930s–1940s to under 4 seconds after 2000, exhibiting an approximately linear decrease on a logarithmic scale over 75 years (the correlation coefficient, $r = -0.75$)—formal features such as cuts, zooms, and sudden noises trigger the involuntary orienting response, thereby keeping attention on the screen (Kubey & Csikszentmihalyi, 2002). Experimental

evidence shows the cost side of this mechanism: fast pacing (defined by the number of cuts) and arousing content each increase the allocation of processing resources, while their combination overloads the processing system, resulting in less recognition and cued recall for the specific content of the message (Lang et al., 1999). Cutting's sample consists of films; television has its own series in Butler (2014): the annual average shot length of American narrative television programs often exceeded 14 seconds in the 1950s, and by the 2000s never reached 7 seconds (Butler describes his sample collection as haphazard, with visibly fluctuating annual data points). Experimental research sets 11 or more scene changes within 30 seconds as fast pacing (Lang et al., 1999). Within the same counting convention, the film and television series each show a fewfold shortening in pacing—down to roughly forty to fifty percent of the original (this book's derivation), not a jump in order of magnitude. Postman's (1985/2006) description of "amusing ourselves to death"—public discourse degrading from the argumentative to the fragmentary—is a cultural symptom of this transition zone.

The essence of the transition zone is not how many times the raw density of the media conductivity coefficient has increased—it is that control has begun to migrate from the host to the projecting party. Viewers cannot control the broadcast rate; advertising embedding creates compulsory projection. But television still has physical constraints: a limited number of channels, fixed broadcast time slots, and a low decision cost for the act of turning it off.

Television advertising pushed the migration of control further. Single advertisements on American network television grew shorter and denser between 1965 and 1995: average length fell from 53.1 seconds to 25.4 seconds, and the number of ads per minute rose from 1.1 to 2.4 (Potter, 2004). The terms attached to an advertisement can be read only within broadcast time: in the Canal Digital case the Court of Justice of the European Union pointed out that a newspaper reader has sufficient time to read an integral advertisement, whereas the time available to the television consumer to assess the information provided is limited (Straetmans, 2019). The court's distinguishing of television from print by "insufficient available time" is consistent with this chapter's verification-window dimension; the judgment acknowledges a difference in assessment time between the two media and does not directly establish the audit-pressure ratio of this scenario.

Television advertising is but one species of advertising-class arrival. Such arrivals share a cost structure: the costs of processing, discarding, and being interrupted fall on the recipient, not the marketer (Petty, 2000 calls them consumer-borne marketing costs); the cheaper the sending, the more willing marketers are to send messages to people unlikely to be interested, and the market therefore systematically oversends. When arrival is excessive, the recipient's response is not to process: fast-forwarding through advertisements on videotape, barely noticing magazine advertisements, discarding direct mail unopened, screening telemarketing calls, and not opening most Internet advertisements (Petty, 2000, citing Godin, 1999); about half of all direct mail is disposed of without examination (Petty, 2000, citing Headden, 1997). In 1998 the American credit-card industry sent 3.4 billion solicitations by mail, nearly four times as many as in 1992, while the response rate slipped to 1.2% of the offers, down from 2.8% in 1992 (Petty, 2000, citing Sanders, 1999). Petty further notes that interactive media require the recipient to operate even if only to bypass the advertisement, so the costs imposed on recipients without their consent only increase. This is the face of "passing without audit" when the ratio exceeds 1.

Subtitles are the scenario of "the medium sets the rhythm, but calibrates to the viewer": subtitles appear and vanish with the picture, and the viewer cannot adjust the pace, yet their speed norms are set to the viewer's reading ability. Conventional speeds have risen from 12 characters per second to 17–20 cps (Szarkowska &

Gerber-Morón, 2018); converted at 5 characters per word (the convention of Zheng & Meister, 2025), the adult silent-reading speed of about 238 words per minute for non-fiction amounts to about 20 characters per second (Brysbaert, 2019; this book's derivation)—subtitle arrival and silent reading coincide at about 20 characters per second, a ratio of about 1. In experiments, viewers' comprehension showed no difference at 12, 16, or 20 cps (Szarkowska & Gerber-Morón, 2018); beyond 20 cps, viewers shifted from word-by-word reading to coarser skimming (as cited in Szarkowska et al., 2024). The speed at which viewers shift to skimming is close to the position where the ratio reckoned at silent-reading speed is about 1, but this correspondence depends on taking 238 words per minute (≈20 characters per second) as the reference; what the experiments directly support is that comprehension shows no significant difference at 12, 16, or 20 characters per second and that reading shifts to skimming at higher speeds—not that a verification threshold falls exactly at ratio 1. Subtitles also show that "who sets the rhythm" and "whether the speed is calibrated to the host" are two independent dimensions.

Layer-by-layer superposition of dimensions from television to AI. The move from the transition zone to the turbulent zone is not a single abrupt change—it is a cumulative process: each generation of computer-based media, building upon the previous generation, eliminates one more stop signal and opens up one more new dimension of the media conductivity coefficient. The following table, generation by generation, inventories the migration path of control from the host to the projecting party (the datings mark representative products/events):

Table 3. Control Dimensions and Stop Signals in the Generational Superposition of Media

New Dimension of the Media Conductivity Coefficient	Representative Milestone	Representative Medium	Structural Consequence
Interactivity + Variable-Ratio Reinforcement	1970s	Standalone Games	Feedback loop; built-in stop signals such as save points exist
Hypertext Branching	1991	Web	Choice itself becomes a cognitive burden; the host still controls the initiation right
Persistent World + Social Obligation	1997	Online Games	Stop signal removed (world does not pause); the only behavioral addiction whose diagnostic object is a specific medium-use behavior (gaming) to be formally included by the World Health Organization (WHO) (WHO, 2025)

New Dimension of the Media Conductivity Coefficient	Representative Milestone	Representative Medium	Structural Consequence
Push / Always-On	2000s	Mobile Phones	Merely receiving a notification causes attentional disruption comparable in magnitude to actively using the phone (Stothart et al., 2015)
Algorithmic Sorting + Social Proof	2006	Social Media	Media conductivity coefficient shifts from Pull to Push; manual refresh required (manual initiation is itself a stop signal)
Algorithmic Personalization + Infinite Scroll	2016	Short Videos	Stop signal engineered out (Monge Roffarello et al., 2023; Ye, 2025); the media conductivity coefficient becomes a function of the host's psychological vulnerabilities
Semantic Trust + Cognitive Offloading	2022	AI-Generated	Default acceptance may inhibit the initiation of checking and may correspondingly reduce checking work (a mechanism hypothesis; see §2.2); from cognitive offloading to will-offloading (Wood, 2026)

This table is more important than any numerical value of the cognitive Reynolds number. It demonstrates a direction that has shown no reversal to date: *each generation of media, building upon the previous generation, eliminates one opportunity for the host to stop, think, and check.* The migration of control is unidirectional—no generation of commercial media, after becoming the mainstream medium, has systematically returned control to the host. There are cases of partial restitution (such as platforms restoring chronological-timeline options), but the default arrangements show no systematic return of control—the unidirectionality judgment is confined to the level of default arrangements.

The World Wide Web placed terms behind the "agree" button. A coding of 45 privacy policies among the hundred most-visited websites globally in 2012 showed policies averaging about 2,400 words; not one presented

the policy by default at registration, and only about 18% included a link to the policy when asking the user to check agreement (Steinfeld, 2016). The default arrangement is set by the projecting party; whether to read looks like the host's decision, yet what the host faces is a daily reading load of about 40 minutes (§2.2). The same study's eye-tracking experiment (128 undergraduate participants) measured how default settings change actual reading investment, not behavior when the ratio exceeds 1: when agreement could be checked without reading, only 13 of 64 participants opened the policy, and this group devoted an average of only about 5 seconds to it; when the policy was presented by default, participants dwelt about 59 seconds on average and also understood their rights better (Steinfeld, 2016). (The eye-tracking experiment used a shorter 451-word sample policy, not the 2,400-word site average cited above.) At silent-reading speed, reading the full policy takes about 1.9 minutes (this book's derivation): the non-default group's average dwell is about a twentieth of what reading the full text requires, the default group's about half. These proportions characterize actual investment under given defaults, not the ceiling of the host's available audit time, and they do not by themselves prove that the scenario's ratio exceeds 1—the ratio is computed, per the caliber of §2.2, as verification time required against time available, whereas what the experiment measures is behavior. A user who agrees without knowing the terms may unknowingly consent to uses of personal information she does not approve of (Steinfeld, 2016)—the "agree" button lets unaudited terms take effect directly.

The effect of interface defaults on this behavior has now been quantified experimentally. A field experiment with 72,208 visitors showed that with consent pop-ups identical in content but differing only in default setting, 86.72% of users rejected the additional cookies when they were deselected by default, whereas only 5.60% rejected—and 93.43% accepted them all—when they were preselected (SERNAC, 2022); removing the "reject all" button from the first screen raised the consent rate by 22–23 percentage points (Nouwens et al., 2020). Acceptance on the arrival side is one click; rejection on the defense side is multi-step navigation—the effect of defaults is reckoned in tens of percentage points, and the distribution of control over reception is thereby measured.

Social media further amplify audit-awaiting arrival through retweet networks: a Twitter diffusion study that used the judgments of six independent fact-checking organizations as its standard found that, in all categories of information, falsehood diffused significantly farther, faster, deeper, and more broadly than the truth, and that it took the truth about six times as long as falsehood to reach 1,500 people (Vosoughi et al., 2018)—the study compares post-hoc truth classifications and does not directly identify whether forwarders verified an item at the time. The number of items awaiting audit depends not only on the host's chosen sphere of attention but also on forwarding and recommendation structures. Release timing is itself a scheduled dimension: addiction-optimization practitioners, interviewed on television, described the practice of withholding some likes and releasing them in a burst when the algorithm predicts the user may leave—"for this user right now who is experimental subject 79B3 in experiment 231, we think we can see an improvement in his behavior if you give it to him in this burst instead of that burst" (CBS, 2017).

The "Structural Consequence" column in the above table mixes three types of information of different natures: (a) **objective descriptions** of the design features of the medium (e.g., "stop signal engineered out" describes the interface design of infinite scrolling and autoplay itself, which are objectively verifiable facts); (b) **structural inferences** about the possible effects on host behavior based on design features (e.g., "the trigger for checking may come under strong pressure and checking work may be compressed" is an inference about behavioral consequences under those design features); (c) **cognitive effects** that have already received preliminary support

in the empirical literature (e.g., "merely receiving a notification causes attentional disruption comparable in magnitude to actively using the phone," Stothart et al., 2015). To preserve the readability of the table, the three types of information are not annotated by layer within the table; the reader may refer to the behavioral and neuroimaging evidence in §3.3 to assist in judging the empirical support strength of each entry.

The dimensional superposition table records, from the host end, the generation-by-generation elimination of stop signals; as its dual, the projecting party's formal engineering deepened layer by layer over the same period. In the era of newspapers, radio, and television, the formal engineering was the design of advertising placement and time slots—page position, prime time, insertion rhythm—intervening in the **container** of content presentation. In the era of search and algorithmic distribution, it was Search Engine Optimization (SEO)—optimizing keywords and link structures for ranking functions—intervening in the **visibility** of content. In the era of generative AI, Generative Engine Optimization (GEO) has emerged—optimizing the citability and structure of content for the ways in which models synthesize and reformulate information (Aggarwal et al., 2024)—intervening in the very **manner of representation** of content: the projecting party no longer needs the host to see the original content; it needs only the generative engine to reformulate it on its behalf. From container to ranking to representation, the projecting party's intervention deepens layer by layer—forming the obverse of the generation-by-generation elimination of stop signals, the projecting-end record of the same migration of control.

The above is the generational record viewed from the perspective of the projecting party. Viewed from the perspective of the defense side, the same direction appears in a symmetrical form: the projecting party, through changes in the underlying architecture, systematically marginalizes the operating environment of each generation of counter-tools—narrowing their coverage or withdrawing the interfaces on which they depend.

Counter-Mechanisms: Their Emergence and How They Were Leapfrogged. Unidirectional migration does not mean the defense side has never attempted to reclaim control. Counter-tools appeared at each stage—but the projecting party rendered each generation of counter-tools ineffective at a higher level of abstraction:

- **Digital Video Recorders (DVR) and Ad-Skipping (1990s–2000s).** DVRs allowed viewers to skip television ads—Wilbur (2008) analyzed how this technology destabilized the traditional television advertising model. The projecting party's response went beyond fixing ads: it was to move advertising out of skippable breaks and embed it in content that could not be skipped separately—product placement, native advertising, brand-sponsored storylines. Streaming changed the technical conditions of viewing and advertising control; the scope of the traditional DVR contracted accordingly, with some functions migrating inside platforms.
- **Really Simple Syndication (RSS) Readers and the Open Web (2004–2013).** RSS allowed users to subscribe to sources and arrange reading in an aggregator, sort it chronologically, and read it without ads—almost perfectly preserving control of the media conductivity coefficient in the hands of the host. Its control function depended on sources continuing to provide open subscriptions; when information moved to closed platforms that offered no corresponding interface, the reach of RSS was constrained. The shutdown of Google Reader (July 1, 2013) is a symbolic node, but it does not mean RSS as a whole became unusable—Google's announcement at the time explicitly mentioned RSS alternative tools and subscription export (Hölzle, 2013).
- **Browser Ad-Blockers and the App Ecosystem (2010s–).** Ad-blockers remain widely used on desktop browsers—approximately 30% of global internet users use ad-blocking tools at least occasionally (Glob-

alWebIndex (GWI) data for Q2 2025: 29.5%, an estimated 1.77 billion users; cited in Backlinko, 2026). A distinction is needed: the proportion of use is not the same as blocking efficacy, and the environment a browser plugin can intervene in is not the same as the information streams of all mobile applications. And the structural migration of user activity from browsers to apps **marginalized the operating environment of blockers**: information streams within apps do not pass through the browser rendering pipeline; blocking plugins cannot intervene. The tool itself did not fail—the battlefield on which it operated was moved. The projecting party, by changing the technological substrate—from open protocols to closed apps—caused **browser-plugin-based defense tools to systematically lose their operating environment in mainstream mobile usage scenarios**.

- **Third-Party Clients and Platform Application Programming Interface (API) Lockdown (2010s–2023).** Third-party Twitter clients once offered users a chronological timeline and an experience free of algorithmic recommendations and ads—the latter not because the clients actively stripped out advertising, but because the platform did not serve ads through the API (Clark, 2023). In early 2023, X/Twitter cut off third-party clients' API access and changed its rules to bar apps that competed with its own—Twitterrific was discontinued, Fenix was pulled from app stores, and Tweetbot ceased service (Clark, 2023); free academic API access was terminated thereafter (Blakey, 2024)—this case shows that a platform can, by reclaiming interface access, directly restrict the ways in which users arrange reception through third-party tools.
- **Screen Time Controls (2018–).** Apple and Google's screen time tools elevated the line of defense to the operating system level. But these tools require manual setup, and under a restriction model that users set themselves and are allowed to extend themselves, the effect of control still depends on user execution. The platform side, in turn, counters with interface design: video platforms hide elapsed viewing time, weakening users' autonomy in monitoring how much time they have spent; social platforms lengthen the exit and deactivation flow, ending it with a pop-up that urges the user to log out rather than deactivate the account, while any subsequent login automatically reactivates a deactivated account (Monge Roffarello et al., 2023).

The structural pattern is clear: **defense tools upgrade layer by layer (hardware buttons → software plugins → operating system settings), while the projecting party responds layer by layer with changes in the underlying architecture (skippable ads → unskippable placements, open Web → closed apps, public APIs → locked-down APIs), causing the operating environment of defense tools to be systematically marginalized.** These counter-mechanisms did not appear simultaneously—they were each born into different technological strata (DVR 1990s, RSS 2004, browser ad-blockers 2010s, third-party API clients 2010s, screen time controls 2018) and were respectively leapfrogged by the architectural shifts of the projecting party in the corresponding strata. The record of counter-mechanisms being constrained layer by layer is consistent with the unidirectional proposition; whether the migration is unidirectional overall requires a specified scenario, time window, and net-control metric: it is not that no one tried to reverse it; it is that each attempt at reversal was swallowed by a deeper architectural shift at each level. This directional judgment does not depend on the precise numerical value of the cognitive Reynolds number.

Cross-study data compiled by Gloria Mark shows that screen attention duration declined from approximately 150 seconds (2004) through approximately 75 seconds (2012) to approximately 47 seconds in 2016–2021 (range 44–50 seconds; two independent replications give the same order of magnitude: an average of 50 seconds for

twenty software developers and 44 seconds for fifty office workers, both as cited in Mark, 2023). Due to heterogeneity in measurement methods and samples, this sequence should be regarded as suggestive rather than confirmatory evidence. A distinction between two quantities is needed here: what Mark measures is the attention the host actually devotes to a single item—the object measured by the §7 protocol—whereas the numerator of the §2.2 ratio takes the due time required to audit one item, independent of how much the host actually invests (§2.2). The two quantities move in opposite directions: the due time does not shrink as the medium gets shorter, yet actual investment has collapsed by nearly two-thirds over the seventeen years from 2004 to 2021—this sequence does not jointly measure verification demand and the audit budget and therefore does not directly establish a rise in the ratio; it is compatible with the hypothesis of constrained verification conditions, and whether unchecked commitments accumulate must be tested through the temporal records specified in §7.

§3.3 The Turbulent Interval: Convergence of Three Types of Evidence

The qualitative transition from laminar flow to turbulence does not depend on precise bits/s estimates of the media conductivity coefficient for each stage. It is supported by the convergence of three types of evidence (the independence of the three methodological traditions is limited; see later in this section).

Behavioral Evidence. The duration of individual videos on short-video platforms is mostly within a few seconds to 60 seconds (Jain et al., 2025); average daily usage time has risen rapidly—TikTok's average daily use increased from 27.4 minutes in 2019 to 55.8 minutes in 2023 (cited in Jain et al., 2025; the original data are Oberlo commercial statistics), with the mean for a Greek convenience sample of 18–19-year-old adolescents (sample size $N = 219$, 81.3% female; Bilali et al., 2025) reaching approximately 2.77 hours/day—the gender and national composition of that sample limit its generalization. Infinite scrolling eliminates the stop signal; autoplay eliminates the decision cost (Monge Roffarello et al., 2023; Ye, 2025): the verification window for each item is compressed to the level of seconds, making independent verification, reflection, and judgment potentially harder to undertake—short videos are first estimated by converting their output rather than by substituting item duration for verification time. Transcribed at a normal speech rate of 150–200 words per minute, one hour of continuous speech corresponds to 9,000–12,000 words; at ordinary reading of 238 words per minute, the reading time of the transcribed text is about 37.8–50.4 minutes, a reading term of about 0.63–0.84; the numbers, persons' identities, product functions, and action instructions in the visual track are recorded separately as visual assertions, not omitted by the audio-track transcription. The overload pressure lies not in the reading term but in the counting of the compression–restoration verification component and deferability—the checkable propositions carried by each video, converted at their restoration references, exceed the margin left by reading by tens of times (§2.3), and the verification window that closes as the video scrolls away causes the excess to sediment directly. Verification depth is then determined by source and proposition domain—the source–domain coefficient is the expansion of the §2.3 responsible-party coefficient by source and domain, and its parameterization is not unfolded in this chapter. The cognitive basis: credibility judgments are bounded by disciplinary context—students' ascription of credibility to a professor is confined to the subject the professor teaches and withdrawn in other domains (Rieh & Hilligoss, 2008); expert sources and celebrity endorsements also differ in persuasive effect in advertising experiments (Wilson & Sherrell, 1993, a meta-analysis of 114 experiments; Dean & Biswas, 2001; as cited in Armstrong, 2010). These studies measure credibility judgments and persuasive effects, not source reliability or verification hours themselves; the source coefficient is therefore a scenario parameter—used first for estimation, then calibrated against evidence-tracing and verification records. On this basis, the editorial processes of formal media lower part of the routine verification cost only when the domain

matches and the evidence is transparent, and personal renown does not transfer across domains; the source coefficient should be understood as the residual verification burden borne by the host after upstream calibration has been covered, then calibrated with real output and verification records, never halting the estimate merely for want of an existing measurement. This chapter has no direct measurement of verification time, so whether the ratio exceeds 1 is still read from the scenario interval above, not written as a definite number; whether audit headroom is thereby squeezed is a mechanism claim with no direct measurement as yet (see the distinction between levels of convergence at the end of this section).

This engineering is not unique to the digital age. Courtwright (2019) uses the term "limbic capitalism" to trace a longer historical thread: commercial systems continually target rapid human reactions and suppress dispassionate thinking; he summarizes this commercial logic as "addiction by design" (a phrase borrowed from Natasha Dow Schüll's study of the same title, *Addiction by Design*, as Courtwright notes)—short videos are merely the latest form of this lineage in the algorithmic age (this is cited only as a historical–political-economic framework; its "limbic system/prefrontal cortex" dichotomy is a metaphorical simplification, not neural-mechanism evidence). Paulus et al.'s (2023) critical review reports that most meta-analyses find associations between screen media activity and adolescent anxiety and depression, but that the effects are generally small, that other studies find no or a negligible relationship, and that the overall relationship is complex.

E-commerce and livestream shopping are special forms within the turbulent interval: the core mechanism is not sensory overload but the manipulation of decision architecture (this chapter takes their decision-architecture dimension: time scarcity and interaction design may compress the checking conditions before purchase decisions—impulsive buying as such does not equal a turbulence verdict; see the two-type distinction in §6.3)—the time scarcity of livestreaming increases impulsive buying, with the mediating pathway through perceived urgency holding for utilitarian products (Hao & Huang, 2023), and the streamer's interaction orientation induces impulsive buying by way of emotional arousal and cognitive trust (Wang et al., 2025).

Neuroimaging Evidence. Su et al. (2021), using fMRI, compared viewing TikTok's personalized recommendation videos with generalized recommendation videos shown to new users, and found higher activation in the DMN sub-components and the ventral tegmental area (VTA) under the former condition; the VTA showed significant activation only for personalized videos and not for generalized ones—the paper interprets this selective activation as personalized videos carrying higher value-based representations, and the VTA is a key node of the reward and addiction circuitry. Yan et al. (2024), using electroencephalography (EEG; 48 participants recruited, 45 included in the EEG analysis), found that short-video addiction tendency showed no significant correlation with any of the three behavioral efficiency indices of the Attention Network Test, nor with the EEG indices reflecting the alerting, orienting, and executive-control networks; the only significant negative correlation appeared in the prefrontal theta-power difference between the incongruent and neutral conditions under task conditions requiring executive control (no such correlation was found for resting-state EEG)—the paper accordingly infers that an increased addiction tendency may impair executive control, though no corresponding significant association was found on the behavioral measures used. Loh and Kanai (2014) found that the higher the level of media multitasking activity, the lower the gray matter density in the anterior cingulate cortex (ACC) (a cross-sectional correlation in which the direction of causality cannot be determined)—a region closely associated with cognitive control and emotion regulation.

Clinical Diagnostic and Epidemiological Evidence. The eleventh edition of the *International Classification of Diseases* (ICD-11) entered into effect in 2022; its 2025-01 version classifies gaming disorder under disorders

due to addictive behaviours—the only behavioral addiction formally included by the WHO whose diagnostic object is a specific medium-use behavior (gaming) (not excluding gambling disorder, listed alongside in the same group; WHO, 2025). This is the only entry in this chain of evidence that reaches the level of a clinical entity. The remaining data belong to epidemiological evidence: Sohn et al.'s (2019) meta-analysis targeting children and young people (CYP) (41 studies, sample size $N = 41,871$) shows a median prevalence of problematic smartphone use of 23.3%; Al-Mamun et al.'s (2025) meta-analysis (43 studies, sample size $N = 36,656$) shows that among those tested, 26% had mild, 51% moderate, and 21% severe nomophobia symptoms. These data are prevalence estimates based on research scales, not confirmations of clinical entities—but their directional consistency is evident; the prevalence figures are limited by scale-based measurement criteria and cannot be read directly as clinical prevalence rates. Kubey and Csikszentmihalyi (2002), using the experience sampling method, argued that "television addiction is no mere metaphor"—a claim already advanced in the television era, when the cognitive Reynolds number was merely in the transition zone.

The three types of evidence come from different methodological traditions: behavioral data are aggregated from usage statistics; neuroimaging is directly measured in the laboratory using fMRI and EEG; clinical diagnosis proceeds through psychiatric assessment procedures. Each type of evidence has its own limitations—behavioral data rely on self-reports and platform statistics, neuroimaging samples are small and experimental tasks simplified, clinical diagnoses involve cultural differences and comorbidity interference—but the differences among these methodological traditions make convergence a powerful inferential strategy. This independence, however, is limited: the review source of the behavioral chain (Jain et al., 2025) itself includes and discusses the neuroimaging study by Su et al. (2021), so the three chains are not fully separate.

At the meta-analytic level the behavioral chain splits three ways: a twenty-year meta-analysis of the limited-capacity tradition of mediated message processing (142 articles, 683 effects) found a pooled effect of r = .326 for the cognitive load domain, with no publication bias detected (Huskey et al., 2020)—aggregate support for the capacity-limitation premise; a meta-analysis of 49 media-multitasking studies found negative effects on cognitive outcomes, with the effect larger the less control users had over the pace and sequence of the information flow (Jeong & Hwang, 2016)—meta-analytic moderator evidence for the control-over-reception dimension; and the association between media multitasking and distractibility, in two high-powered replication studies (five of fourteen tests significant, only two surviving a conservative Bayesian analysis) and a meta-analysis of 39 effect sizes, proved weakly significant and nonsignificant after correction for small-study effects, leading its authors to question its existence in laboratory tasks (Wiradhany & Nieuwenstein, 2017).

Read together: capacity limitation and the cognitive costs of multitasking enjoy aggregate support, whereas the individual-differences hypothesis that heavy multitaskers filter worse has not survived replication—consistent with this chapter's layered judgment that directional convergence is strong while the mechanism and individual levels remain indirect.

A distinction between two different levels of convergence is needed here. The three types of evidence, from different methodological traditions, jointly point to the conclusion that "the media environment from short videos onward shows mostly negative associations with attention, executive control, and mental health (with generally small effects, and some studies finding no association—see Paulus in the preceding paragraph)"—this **directional convergence is strong**. But their convergence on the **specific mechanism** of this association—that the information arrival rate has surpassed the conscious processing bottleneck of 10 bits/s—is **indirect**: no single type of evidence directly measures the actual information processing rate of the host during media use. The

inequality between arrival and processing has a formal-modeling precedent in system dynamics: Pavlov et al. (2008) constructed a model of message-based communication in which the information-processing capacity of recipients is limited, with information overload occurring precisely when more messages arrive than recipients can process (the model is denominated in message counts, not a bits/s rate measurement). This chapter uses "the cognitive Reynolds number entering the turbulent interval" as a conceptual tool to explain this mechanism, but the tool itself is not directly derived from these three types of evidence.

Under this premise, the three converge on the same judgment: **in the media environment from short videos onward, the rate and compulsoriness of information arrival conflict with the order-of-magnitude constraint of human conscious cognitive processing (especially that portion of processing capacity required for meta-audit)—the absolute-time compression of the short-video type has direct behavioral evidence, whereas the mechanism claim of rate transgression is compatible with the three types of evidence but has no direct measurement as yet.** This judgment does not require the support of a precise numerical value for the cognitive Reynolds number—it comes from the directional convergence of the three types of evidence (which remains indirect at the mechanistic level; see the preceding paragraph).

The domain of financial trading provides another extreme instance of the order-of-magnitude gap between the projection-end clock and the biological clock. In MacKenzie's (2021) fieldwork, an interviewee reported response times in high-frequency trading (HFT) as low as 42 nanoseconds:

> In March 2019, an interviewee told me that, although his own systems were slower than this, he had learned of the achievement of response times as low as 42 nanoseconds. (MacKenzie, 2021)

The minimum magnitude of human simple reaction time is about 140–180 milliseconds (about 180 for visual stimuli, about 140 for auditory; Luce, 1986), and taking about 200 milliseconds as an illustrative baseline is already a generous estimate—this baseline is roughly 4.8 million times the HFT latency reported by the interviewee (200 ms / 42 ns $\approx 4.8 \times 10^6$). This is not a diagnosis of the cognitive Reynolds number ($Re_c = \Lambda / E_c$)—in high-frequency trading, the human host has been removed from the real-time execution loop; the "decision-maker" is the algorithm. What this comparison shows is the gap of roughly 4.8×10^6 between HFT system response latency and human simple reaction time—it is not a measurement of the ratio of Chapter 3's techno-capital clock (τ_c) to the governance clock (τ_g). This is a footnote on the physical limit of the unilateral speed-up of the projection-end clock.

The deeper structural judgment is:

> That nanoseconds are important in HFT makes its world Einsteinian: for HFT, that no signal can travel faster than the speed of light in a vacuum is a practical constraint, not just a theoretical limit. (MacKenzie, 2021)

This nanosecond-level physical limit, with the speed of light in a vacuum as its actual constraint, is a footnote on the physical limit of the unilateral speed-up of the projection-end clock (for its relation to the cognitive Reynolds number (Re_c), see the declaration at the head of the HFT passage in this section); it does not constitute a diagnostic that "the techno-capital clock has already been pushed to the point where only physical laws can constrain it"—the latter being the argumentative domain of Chapter 3 itself, referenced here only as a conditional comparison. The implication for the cognitive Reynolds number framework is indirect but directionally clear: when the physical limit of the projection-end clock has been pushed to the constraint of the speed of light, while

human reaction time still remains on the order of hundreds of milliseconds, the rate asymmetry between the two has reached such a magnitude that humans have been removed from the real-time execution loop—the qualitative change lies in this structural fact, not in the numerical ratio itself.

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§4 The Age of AI Generation: Why the Simplified Form of the Cognitive Reynolds Number (Re_c) Is No Longer Sufficient

The analysis of the first three intervals used the two-variable simplified form introduced in §1.1. This simplification is justified only when unlocked stock, narrative load, meta-audit capacity, and the legitimacy base can be treated as background conditions and absorbed into the scenario scale within a specified short time slice; cross-period or macro-level analysis cannot retain that simplification (Chapter 3, §5.2).

A declaration cutting the ordinal from the cardinal must be made explicit here: the cardinal claims of this chapter are of only two kinds—ratios computed in the same unit within specific scenarios (§2.2), and time series of the same construct under the same counting convention (such as average shot length or subtitle speed)—and the within-scenario ratio is computed only within short time slices of the simplified two-variable form (Chapter 3, §5.2). Every leap made in what follows and by the complete AI form (the trigger for checking possibly coming under strong pressure, checking work possibly being compressed, personalization precision, migration of control) **bears ordinal judgment only** (which is higher, in what direction) and no longer claims any cardinal value carrying a unit. This distinction demarcates three boundaries: the cardinal orders of magnitude of the simplified form, the ordinal direction of the complete form, and the fact that the two must not be mixed.

In the AI age, this condition no longer holds. The legitimacy base ($\|L\|$) receives no independent dynamic claim here: neither its within-stage constancy nor its absorption by proxies is established, so the folding condition of the preceding paragraph remains for it an untested premise rather than a demonstrated fact; variation in the legitimacy base ($\|L\|$) within particular AI scenarios must be examined separately. The structural position of the three variables has changed fundamentally—the unlocked stock (C) has independent exponential-growth data, while the changes in narrative load (ϕ) and meta-audit capacity (A) can so far only be characterized qualitatively—they can no longer be folded into background constants:

The following discussion of unlocked stock (C), narrative load (ϕ), and meta-audit capacity (A) is a **qualitative diagnosis**—this chapter does not provide quantitative proxy indicators for them, nor does it possess the order-of-magnitude positioning directly anchored by a neuroscience review as §2.1 does for the human biological substrate (E_c). The complete form of the cognitive Reynolds number (Re_c^{AI}) (given at the end of this section) provides the structural outline for Part II (the AI application chapters: Chapters 9, 11, 13, and 15); the operationalization and empirical anchoring of proxy indicators belong to the work there.

- **The unlocked stock (C) is no longer a background constant.** According to estimates by research institutions such as Epoch AI, based on publicly disclosed data, global AI computing power roughly doubles every 7 months (You et al., 2026); the Stanford Institute for Human-Centered Artificial Intelligence (Stanford HAI) AI Index report of 2026 estimates that global AI computing power has grown at approximately 3.3×/year since 2022, reaching approximately 17.1 million H100-equivalent compute units as of 2025—a secondary report within the same data chain; and training computation has roughly doubled every 6 months since 2010 (Rahman & Owen, 2024)—these are trend estimates, not precise measurements, but

the directional order of magnitude is clear. The unlocked stock itself is growing exponentially—it is not an appendage of the media conductivity coefficient but an independent variable that amplifies synergistically with it. What needs to be annotated is: the unlocked stock (C) does not directly act on the host—a trillion-parameter model and a ten-billion-parameter model, arriving at the host through the same screen and the same API, can have the same bits per unit time. The mediating mechanism by which the unlocked stock (C) enters the numerator of the complete form is: higher unlocked stock (C) increases the real-time adaptive precision of the narrative load (ϕ) and raises the personalization precision dimension of the media conductivity coefficient (Λ). That is to say, the unlocked stock (C) is a driving variable of the media conductivity coefficient (Λ) and narrative load (ϕ), not an additive contribution—the complete form lists it separately to mark its independent dynamics in the AI age, which can no longer be folded into other variables, not to claim that it is statistically independent of the media conductivity coefficient (Λ) and narrative load (ϕ). The multiplicative structure of the complete form accordingly requires the three indicators to each cover non-overlapping dimensions: the unlocked stock (C) counts deployment scale, the media conductivity coefficient (Λ) counts per-unit arrival efficiency (with scale stripped out), and narrative load (ϕ) counts the narrative effect per unit of output—the values of the media conductivity coefficient (Λ) and narrative load (ϕ) must be understood as unit intensities with scale stripped out.

- **The narrative load (ϕ) is no longer static camouflage.** The output of Large Language Models (LLMs) is fluent in language and naturally coherent—what is fluent is the appearance, not the reliability of the content; Chapter 9, §4 argues for a conditional mechanism: when annotators judge preferences from single outputs without independent verification, and the reward cannot distinguish factual correctness from ingratiating presentation, Reinforcement Learning from Human Feedback (RLHF) can produce pseudo-small fibers—the model may learn to present uncertain content in a tone of certainty. The narrative load evolves from "static information asymmetry" to a "real-time generated, adaptively aligned" narrative. The concept of "will-offloading" proposed by Wood (2026) points in a direction worth noting (Wood takes autonomous and AI-enabled systems in the military domain as his example and states that the underlying arguments extend to such systems more generally; his distinction is bidirectional—systems used in a "will-extending" manner can remain under meaningful human control even when operating outside the direct control loop, and this book takes only the "will-offloading" side as a risk hypothesis, applying it further to ordinary users, a step that is this book's own structural analogy): the issue is not that users cannot audit, but that they gradually become unwilling to do so. The empirical basis for this judgment still needs further accumulation. The mechanism hypothesis proposed here is that the effect of narrative load is not limited to concealing information but may also reduce the frequency with which checking is initiated. This hypothesis is compatible with the chapter's framework, but it requires testing against independent behavioral records.
- **Meta-audit capacity (A) is no longer a constant within the stage.** The two degradation pathways argued in Chapter 3, §4.2—the automation paradox discarding operational friction and algorithmic xiangyuan displacing affective friction—accelerate simultaneously in the AI age. The decay dynamics of meta-audit capacity itself becomes a process requiring independent tracking.

Therefore, in the AI age, one must return to the complete form (the general form is given in Chapter 3, §5.2; here it is instantiated for the AI scenario):

$$Re_c^{AI} = \frac{C \cdot \Lambda \cdot \phi}{A \cdot E_c \cdot \|L\|}$$

The cognitive Reynolds number in its complete form (Re_c^{AI}) no longer retains the intuitive physical analogy with the fluid Reynolds number. It is best treated as the **ordinal ratio between "total projection-end pressure" and "total defense-end capacity,"** and no longer force it into the mechanical imagery of "inertia/viscosity." The retention of the "Reynolds" naming here is a nominal inheritance from the simplified form and does not carry the explanatory function of the physical analogy—the "laminar/turbulent" metaphor of the simplified form should not be extrapolated to the complete form. This expression is a **classificatory structural expression**—inheriting the classificatory discipline of the Chapter 3 meta-formula: it marks which factors sit in the numerator, which in the denominator, and their directional assumptions; it is not used for numerical computation. The operationalization of the complete form will be unfolded in the AI application chapters of Part II—the task of this chapter is to establish the applicability boundary of the simplified form and the necessity of the complete form.

The complete form is not used for numerical computation; but within a short time slice of the same scenario, the simplified form can still be reckoned as in §2.2–§2.3—and the AI scenario is easier to reckon than the others: the text generated and the citations and numbers within it all remain in the working record, so arrival volume and actual investment can be checked directly against the log; the work-hours required to complete each verification must still be estimated separately under the definitions of §2.3—actual time spent in the log is not automatically the time verification requires. Generation volume can multiply with generation speed and the number of parallel sessions, while the time available for auditing is constrained by the human biological substrate and does not grow with it; the within-scenario ratio therefore rises with generation volume. If the host persistently says only "continue" without verifying, unverified content becomes the premise of subsequent modification—precisely the starting point of the turbulent-type debt trajectory described in §7. AI output is first estimated by textualization: let the output quantity (G) be the number of visible tokens entering the processing scope within the one-hour window, and calibrate a character-conversion coefficient (α, the average characters per token) on output samples of the current model and language; the text quantity is then $G\alpha$. The generation volume takes the range of 5,000–200,000 tokens per hour (light sessions to heavy parallelism; the demonstration case takes $G = 100{,}000$, $\alpha = 4$). Taking English text at 238 words per minute and 5 characters per word (about 1,190 characters per minute) as the reference, ordinary reading time is $G\alpha/1190$ minutes, with a reading term of about 0.28–11.2 (the demonstration case of 5.6 at the midpoint); Chinese output uses Chinese parameters—Chinese silent reading reckoned in characters, taking the median scenario of 320 characters per minute (the range anchored in the next paragraph).

Chinese reading rates must additionally be calibrated by reader and task. Zhang et al. (2024) selected 24 fast and 24 slow readers from 96 valid participants, with rates of 507±73 and 216±35 characters per minute; comprehension accuracy was 93%±5% and 92%±5%, respectively, with no significant difference. Zhang et al. (2020) reported rates of 458 and 299 characters per minute in another experiment, with both groups at 92% accuracy. Zhang et al. (2023) reported 567±51 versus 275±37 characters per minute in Experiment 1 and 655±133 versus 309±28 in Experiment 2. These task-specific results demonstrate substantial individual variation rather than a single population distribution. Eye-movement studies of natural Chinese reading give rates of roughly 300–345 characters per minute (Brysbaert's figure of 260 wpm for Chinese, converted at 1.5 characters per word,

overestimates texts with longer words—an acknowledged caveat in the original)—this chapter takes 320 characters per minute as the median baseline reference; actual estimates should record task type, reader sample, and comprehension accuracy, with sensitivity checks across other defensible rates.

Fast reading alone entails neither failed comprehension nor absent auditing. The fast and slow groups above differed severalfold in rate despite similar comprehension accuracy. Sui et al. (2025), reviewing speed-reading research, find that training can increase speed while often reducing attention to details, logical inquiry, and deep integration. This chapter therefore distinguishes ordinary comprehension, gist extraction, and external verification: faster reading can improve information extraction, but it cannot substitute for item-by-item comparison with independent references. Skimming or reduced rereading are proxies for processing style, not proof that meta-audit capacity has been lost.

As a mere demonstration of the arithmetic: if the visible output entering the scope of processing in one hour is 100,000 tokens and samples yield $\alpha = 4$ characters per token, the text quantity is 400,000 characters, requiring about 336 minutes (about 5.6 hours) at the above English reference—that figure shows the conversion method only; $\alpha = 4$ is a demonstration value, not a universal constant. On top of this are added the verification times determined by source-checking, fact verification, and the source–domain coefficient (§3.3); commitments accepted and retained without verification enter the debt ledger separately and are not replaced by the raw generation volume. Having AI re-check AI's output cannot substitute for calibration—calibrating a clock with a clock is not calibration (Chapter 1, §1.3): agreement among models does not amount to alignment with an independent reference. When tools provide checkable source texts, calculations, or test results, the work-hours they save in search and comparison enter the estimate; the time to read the evidence and make the judgment remains the host's. The "agree" button on the web has already displayed this structure: when one can continue without reading, most do not read (§3.2).

To ground the above asymmetry in a concrete scenario, a typical diagnostic anchor where all three variables are simultaneously activated is given below—generative AI embedded in clinical decision-making. The editorial by Smith et al. (2026) in *Academic Psychiatry* points out that offloading complex clinical tasks to generative AI **can potentially lead to** a range of cognitive biases—overreliance on automated systems, risk of error, loss of previously acquired skills, and others (the editorial notes that this list is drawn from Macnamara et al.); this book takes three of them and groups them, under its own framework naming, as three types of risk that may reinforce one another: **overreliance on automation** (taking AI output as correct by default and forgoing independent verification, corresponding to the dissolution of the willingness to audit by narrative load (ϕ)—RLHF can produce pseudo-small fibers under the specified annotation and verification conditions, allowing a tone of certainty to mask uncertain content); **amplification of error risk** (AI's false positives and false negatives diffuse through the decision chain to more clinical nodes—if additional unlocked stock is used to expand AI deployment across clinical settings, the number of nodes where AI intervenes will increase, expanding the potential exposure surface of a single error; if computing capacity remains concentrated in a small number of nodes, this expansion path does not hold. The direct cause of error propagation remains system coupling, while the role of the unlocked stock (C) here is an **amplifying substrate**—it determines how widely an error, once it occurs, can propagate); **skill atrophy** (after long-term dependence, clinical judgment capacity may undergo structural degradation, corresponding to the decline of meta-audit capacity (A) along the automation paradox pathway of Chapter 3, §4.2). If the three occur together, their possible common constraint is that verification time is compressed by clinical workflows—this awaits testing in clinical settings. This is only a qualitative

diagnostic anchor: the hypothesis that the three types of risk co-occur within the same scenario is worth testing, but the quantification of proxy indicators is left to Part II.

One step of conditional extrapolation: should systems capable of acting in place of the host across multiple domains arise (the artificial general intelligence (AGI) case), the readings would be the actions executed on the host's behalf and their consequences—their count determined by machine execution speed and the number of parallel agents, and the per-item calibration time at least the time to understand the record of one action and judge its consequences against external references; where consequences are irreversible, calibrating after execution is already too late. Under an idealized assumption that every action requires the host's prior item-by-item verification and must nevertheless be executed, two extreme arrangements emerge: pressing the agent's execution speed down within the host's calibration capacity—each step awaiting the host's calibration before executing, so that the agent runs only at human speed—or the ratio exceeding 1, the host abandoning item-by-item calibration and turning to Strategy 4 of Chapter 1, §1.3 (selecting an authoritative clock)—that is, trusting the agent. This paragraph is an extrapolation without measurement, marking only the limiting shape of the ratio structure in the AGI case.

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§5 The Symbiosis-Parasitism Continuum: An Evolutionary Explanation of the Migration of Control

The dimensional superposition table of §3 demonstrated a directional record: control of the media conductivity coefficient migrates unidirectionally from the host to the projecting party. This section interrogates the dynamics of this migration.

We place the history of media within an evolutionary framework: each generation of media, building upon the previous generation, develops new dimensions of the media conductivity coefficient, advancing the migration of control by one step. This process is described using a symbiosis-parasitism continuum:

Table 4. Control, Host Autonomy, and Relationship Types in the Symbiosis-Parasitism Continuum

Control of the Media Conductivity Coefficient	Host Autonomy	Relationship Type	Representative Stage
Full Host Control	Very High	Commensalism (host benefits)	Books
Partial Migration (One-Way Push)	Medium	Commensalism shifting to Neutral	Broadcasting/Television
Persistent World + Social Obligation	Low	Parasitic Tendency	Online Games
Algorithmic Sorting (manual refresh still required)	Very Low	Parasitism	Social Media
Infinite Scroll + Autoplay	Severely Constrained	Strong Parasitism	Short Videos

Control of the Media Conductivity Coefficient	Host Autonomy	Relationship Type	Representative Stage
On-Demand Generation + Default Trust	Severely Constrained	Symbiosis Reversal	AI-Generated

The "Relationship Type" column in this table is a set of **analogy-based labels** based on the distribution of control, not a measurement of the host's net benefit—the net benefit users derive from each medium (entertainment, information, social value) must be assessed separately, and low control does not automatically equal net harm. The "parasitism" labeling follows the terminological convention of Chapter 3, §1 (the criterion of whether cognitive-debt output systematically exceeds the host's processing threshold), not the biological net-benefit criterion; this table prejudges the tendency by the distribution of control, and the debt empirics for each medium belong to the application chapters. A further distinction is needed: "host autonomy" in this table combines dependence structure with exit costs and is a different dimension from "control over reception" in §2.2 (the rights of initiation and pause)—in generative-AI interaction the initiation right usually remains with the user, but the dependence feedback and the possible strong suppression of the trigger for checking (an inference) leave the effective exercise of exit and independent judgment severely constrained: the initiation right remains, while autonomy has been displaced.

The first five rows of the table (Books → Short Videos) outline the **first half**—**the projecting party actively seizing control**: control of the media conductivity coefficient migrates step by step from the host to the projecting party; the host is the passively captured party; what the dimensional superposition table of §3.2 demonstrates is precisely the generational record of this migration. The last row of the table (AI-Generated) marks a causal turning point: the driving force of the **second half** no longer relies solely on the active capture by the projecting party but is compounded by the active solicitation of the host—the migration of control shifts from unilateral seizure to bidirectional lock-in.

The historical depth of this first half can be drawn much longer than digital media. Wu (2016) organized the "commodification of attention" into an industrial history spanning nearly two centuries, beginning with the insight of Day, the founder of the penny press, in 1833:

> What Day understood—more firmly, more clearly than anyone before him—was that while his readers may have thought themselves his customers, they were in fact his product. (Wu, 2016)

Thereafter radio and television established "exchanging free content for attention" as a mintable currency; by the smartphone era, this logic was pushed into every crevice of "time and spaces previously walled off from commercial exploitation." He writes:

> Thus, bit by bit, what was once shocking became normal, until the shape of our lives yielded further and further to the logic of commerce—but gradually enough that we should now find nothing strange about it. (Wu, 2016)

This thread supplies the unidirectional migration of control in this section with the **historical dynamics at the level of industrial organization and political economy**: each generation of media further commodifies the host's not-yet-harvested consciousness, and the scope of harvesting increases overall. Wu also documents the recurring cycle of disenchantment and revolt: most revolts induced only revised terms from the projecting party,

but some produced regulation and industrial reconfiguration—the Paris anti-poster movement brought posting restrictions, and the patent-medicine industry collapsed under truthful-labeling legislation and public disenchantment; these localized reversals, however, did not reverse the overall expansion of attention commodification—mirroring the judgment in §3.2 that counter-mechanisms exist yet are rendered ineffective layer by layer.

Courtwright (2019) gives a complementary diagnosis from another level and over a time span far longer than that of digital media (reaching back millennia). He uses "limbic capitalism" to characterize a class of business systems that persist in encouraging overconsumption and addiction:

> They do so by targeting the limbic system, the part of the brain responsible for feeling and for quick reaction, as distinct from dispassionate thinking. (Courtwright, 2019)

Courtwright summarizes this commercial logic as "addiction by design" (the phrase originates with Schüll; see §3.3). The history he traces far predates the digital age—from tobacco, alcohol, sugar, coffee, and opium to gambling, addictive commerce is not unique to the digital; short videos are merely the latest form of this lineage (echoing the citation in §3.3). It is necessary to reiterate the boundary already declared in §3.3: this is cited only as a historical-political-economic framework, and its "limbic system/prefrontal cortex" dichotomy is a metaphorical simplification, not neural-mechanism evidence.

Wu and Courtwright do not overlap but are stacked one atop the other: Wu depicts the **institutional layer** of attention being unidirectionally transferred as a commodity, answering "who controls attention"; Courtwright reveals the inducement structure of the same process at the **addictive-commerce layer**, answering "who hijacks the reward circuit." The first half of the symbiosis-parasitism continuum is therefore a **dual-layer advance**—the institutional layer and the addictive-commerce layer operating simultaneously—and this is precisely the root of why the migration of control is so robust: it is not a single commercial strategy but a long-term structure in which institutional incentives and business mechanisms that monetize rapid reward responses are superimposed.

Symbiosis reversal is this chapter's name for this causal turning point: the host's relation to the system moves from receiving pre-made content to a dependent form in which requests and real-time responses reinforce each other—seeking, generation, and demand shaping enter the same feedback process (active seeking itself is not new; the telephone and the Web were already active pull; what is new is the feedback structure). Terminological note: in the strict biological context, the "reversal" of a symbiotic relationship usually refers to a change in the direction of benefit (mutualism → commensalism → parasitism), not a change in the attribution of agency. This chapter borrows the word "reversal" to refer to **the flip in causal structure**, not the biological term.

The reason for choosing "symbiosis reversal" rather than "demand internalization" is: "demand internalization" only describes the change on the host end and does not capture the symmetrical flip on the projecting party end, which simultaneously extends from the delivery of prefabricated content to on-demand generation—the "sym-" (together) in "symbiosis reversal" is precisely meant to mark the structural changes occurring simultaneously in both parties. In mutualism, each party derives some benefit; in the "reversed" relationship, the host actively maintains dependence, and the projecting party further reinforces dependence by satisfying it—forming a positive feedback of bidirectional causality (Chapter 5, §3.2).

In the AI age, the relevant change brought by generative AI is not that users actively solicit information for the first time (the telephone, the Web, and RSS are already forms of active pull), but that the system generates and adjusts its responses in real time during interaction—the host's requests and the system's accommodation

can reinforce one another (the bidirectional causality of Chapter 5, §3.2), reducing independent judgment under specified conditions; active use by itself is not evidence of dependence. The projecting party, in turn, extends from the delivery of prefabricated content to on-demand generation, continuing to shape demand through feedback. The "non-human object" argued in Chapter 5 (Hsu), §3.1 here acquires its positioning within evolutionary dynamics:

> Non-humans can be controlled with far greater certitude than humans. (Hsu, 1983)

Generative AI enables non-human objects to generate responses in open-ended semantic interaction and adjust subsequent responses on conversational feedback ("adjust" meaning conditioning on the interaction, not real-time parameter updates), thereby occupying a new position in the lineage of non-human objects; it not only passively serves as a compensatory tool but also, under the optimization objectives and feedback loops set by its developers, reinforces users' repeated demand for itself (the bidirectional causality argued in Chapter 5, §3.2).

From the perspective of delegation, this migration has yet another complementary explanation. The "control paradox" proposed by Di Nucci (2021) points out: the technologies we develop to enhance control often end up weakening it—he calls the illusion of "mistakenly believing that enhancing control requires no cost and has no side effects" the "empowerment illusion": as if delegation and technological progress were a "free lunch" requiring no trade-offs. A difference in mechanism must be noted: Di Nucci's control paradox originates from the host's **voluntary delegation** (actively ceding direct control in order to enhance overall control), whereas the migration of control argued in §3.2 of this chapter is a redistribution of control caused by the projecting party's architectural shifts—the two share a direction but differ in dynamics. Symbiosis reversal is precisely the confluence point of these two mechanisms: the host's voluntary delegation under the "empowerment illusion"—mistakenly believing that ceding control has no cost—opens the channel for the redistribution of control. This paradox is especially salient in AI clinical decision-making scenarios: Di Nucci's (2021) analysis of medical AI echoes the cognitive-offloading risks listed in the Smith et al. (2026) editorial cited in §4—physicians delegate medical judgment to enhance diagnostic control, yet may in reverse undermine it through skill atrophy and overreliance on automation—precisely the risks Smith et al. list as potential, still awaiting clinical evidence.

Along the media trajectory examined in this chapter, the long-term upward pressure on the cognitive Reynolds number is not technologically neutral "progress," but a directional indicator of the projecting party continuously squeezing the defense side's room to operate in niche competition. When the cognitive Reynolds number is in the turbulent interval under the multi-dimensional criteria (arrival persistently exceeding auditable capacity together with constrained reception control, verification window, or checking trigger—not a single rate threshold), the ecological conditions for the defense side to initiate meta-audit are significantly compressed—the root of this compression lies not in the weakness of individual will but in the asymmetric competition itself on the evolutionary scale. Where does the cognitive root of this asymmetry lie? The host's default information processing is **predictive processing** (a neural computational framework in which the brain makes probabilistic predictions about the environment based on long-formed internal models and only further processes prediction errors), not item-by-item analysis (Peelen, Berlot & de Lange, 2024; it needs to be noted: the review by Peelen et al. targets scene and object perception; this chapter extends it from perceptual predictive processing to the predictive processing of the information environment as a whole by structural analogy; empirical evidence is still accumulating). In the laminar interval, the statistical structure of the information environment is relatively stable, and predictive processing is both efficient and energy-saving; by the turbulent interval, the projecting party contin-

uously reshapes the statistical structure of the information environment through algorithms, and the predictive accuracy of the internal model may be systematically lowered—when the change in the statistical structure of the environment outpaces the updating capacity of the internal model and compensatory mechanisms are insufficient, predictive processing may shift from efficient and energy-saving to persistent mismatch.

Research on Chinese reading provides direct evidence for this narrower predictive-processing claim. Zhang et al. (2023) found that both fast and slow readers used predictive information in foveal lexical processing, with differences emerging mainly in parafoveal processing: slow readers relied more on context, had greater difficulty identifying and integrating targets under unpredictable or invalid previews, and were less effective at suppressing irrelevant information. These results support individual variation in the use of prediction during reading, but they measure lexical identification and preview processing, not external verification or meta-audit capacity. Any application of this mechanism to the wider information environment must therefore be tested separately for its effect on the initiation of checking.

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§6 Estimation Boundaries and Interpretive Value

§6.1 Cannot Be Precisely Calculated

The cognitive Reynolds number cannot be calculated as a precise dimensionless number in the physical sense. The multi-dimensionality of the media conductivity coefficient (the possible interactions among the five proxy dimensions), the context-dependence of processing capacity available for audit (the human biological substrate (E_c) remains approximately constant while the effective exercise of meta-audit capacity (A) varies), and the confounding effect of narrative load in the simplified form—the three jointly make precise calculation infeasible under this framework's accounting discipline (Chapter 3, §7.1: the meta-formula is a chart of accounts, not an equation); within a specific scenario, defining the numerator and denominator by the time auditing requires suffices to construct a local index (§2.2–§2.3); it does not automatically possess cross-scenario comparability—cross-scenario comparison requires prior standardization (same host, same task, same information unit, same accounting period, §1.2); when the conditions hold, comparison proceeds in orders of magnitude; otherwise, only the positions of the ratios relative to their capacity boundaries are compared. Any attempt to use the cognitive Reynolds number as a precisely measurable physical quantity would commit a methodological error.

§6.2 Directional Judgments Can Be Made

But judgments about the general direction are stable, based on the following:

- **Laminar Interval (Books → Broadcasting)**: In reading and telephony, the host can keep audit-awaiting arrival within verification capacity; linear broadcasting belongs only to the comprehension-level upper boundary of laminarity and already has transition-zone auditing conditions. Reading is self-paced, telephony permits negotiated pacing, and linear broadcasting follows the broadcaster's schedule (differences in reception control are discussed in §3.1). Empirical anchors: Brysbaert (2019) (reading speed), Eysenck & Brysbaert (2018) (rereading and speech rate), Coupé et al. (2019) (cross-linguistic constancy of the speech information rate).
- **Transition Interval (Film → Social Media)**: Control of the media conductivity coefficient begins to migrate from the host to the projecting party. Stop signals are eliminated layer by layer. The operating

conditions for meta-audit become markedly constrained. Empirical anchors: Cutting et al. (2011) (shortening shot lengths), ICD-11 (WHO, 2025) (gaming disorder diagnosis), the irreversible direction of the dimensional superposition table.

- **Turbulent Interval (Short Videos → AI-Generated)**: Behavioral, neuroimaging, and clinical diagnostic studies provide heterogeneous evidence. The two types of environment make independent meta-audit difficult through different mechanisms: the short-video type compresses the online verification window during continuous playback, whereas the AI type advances the hypothesis that default acceptance may inhibit the initiation of checking and correspondingly reduce checking work (§2.2 and §6.3). The AI hypothesis is not automatically covered by short-video evidence and requires separate testing.

From laminar flow to turbulence—the within-scenario ratio going from what the host can press below 1 (books) to what conditionally exceeds 1 as audit-awaiting item counts or generation volume grow (short videos: exceeding 1 as soon as the verification component, converted at its restoration references, exceeds the margin left by reading; AI: the ratio exceeds 1 when ordinary-reading and verification time exceed the same window; unchecked content that is accepted, retained, and subsequently invoked is recorded separately as debt under §7)—§3.3 and §4 give conditional judgments, not measured constants; this direction holds under two different strengths of robustness. **Weak Reading**: Even if the human biological substrate (E_c) is taken across the entire range of task estimates in the review's Table 1 (about 5–50 bits/s, §2.1), the auditable amount remains a quantity with an upper bound that does not grow with technology, and the direction of the ratio's rise is unchanged. Subtitle speed changes the mode of reading near the position where the ratio, reckoned at silent-reading speed, is about 1 (a correspondence that depends on the choice of reference; §3.2); the ordinary-reading workload of privacy policies alone amounts to more than half of daily Internet time (§2.2)—instances of this direction in concrete scenarios. The rise of the ratio comes mainly from changes in audit-awaiting item counts and rhythm control, not from a multiplication of single-channel speed—same-construct series show changes only of a few-fold magnitude (§6.3). **Strong Reading**: The relative ordering of the various media stages within the Laminar → Transition → Turbulent intervals will not be changed by future revisions to the value of the media conductivity coefficient (Λ)—this robustness depends on the ordinal stability of the proxy indicators, and its evidential requirements are higher than those of the weak reading. The core macro-directional argument of this chapter requires only the weak reading; fine-grained orderings—the generational record in the §3.2 dimensional superposition table and the directional gradient in the §5 continuum—depend on the strong reading, which is an independent hypothesis demanding more evidence. The weak reading supports the macro direction (the direction of the span from books to AI and the judgment that it exceeds the human biological substrate (E_c)); the strong reading supports fine-grained orderings (the relative positions of media stages on dimension-specific criteria—short videos and AI are different mechanism types and are not ranked on a single scale of turbulent intensity; see §6.3)—the finer the claim, the stronger the reading strength required. This is consistent with the Fermi estimation spirit of this chapter.

§6.3 The True Value of the Cognitive Reynolds Number (Re_c)

The value of the cognitive Reynolds number lies not in a pseudo-precise critical value, but in explicitly parameterized scenario estimates and their structural interpretation:

- **Structural Intuition**: The metaphor of "information flow rate / cognitive viscosity" allows non-specialist readers to quickly grasp the speed asymmetry between the projection end and the defense end.

- **A Scale for Cross-Era Comparison**: Quantities of the same construct give testable magnitudes of change—average film shot length fell from about 10 seconds in the 1930s–1940s to under 4 seconds after 2000 (Cutting et al., 2011); the annual average shot length of American narrative television fell from often over 14 seconds in the 1950s to under 7 seconds in the 2000s (Butler, 2014); conventional subtitle speed rose from 12 to 17–20 characters per second (Szarkowska & Gerber-Morón, 2018)—all changes of a fewfold magnitude (this book's derivation). The cross-era rise of the cognitive Reynolds number comes mainly from the growth of audit-awaiting item counts with push, forwarding, and generation, and from arrival rhythm no longer set by the host—not from a multiplication of single-channel speed. (Reiterating the cut in §4: AI's turbulence comes from the possible strong suppression of the trigger for checking (an inference), the parallel expansion of generation volume, and personalization precision, and the complete form here makes an ordinal judgment only.)
- **Locating the Root of the Audit Predicament**: When the cognitive Reynolds number is in the turbulent interval under the multi-dimensional criteria, the verification conditions required for meta-audit have already been structurally compressed to a degree where operation is difficult—The short-video type compresses the online verification window during continuous playback. The AI type advances a different mechanism hypothesis: semantic coherence and default acceptance may reduce the frequency with which checking is initiated, and when checking is not initiated, the corresponding work does not occur. Checking initiation, actual checking work, and meta-audit capacity are recorded separately; failure to initiate is not equated with loss of capacity, and no general tendency toward zero is presupposed. The claim concerns how media environments shape auditing conditions, not an attribution of audit failure to insufficient individual will. Its organizational form has a practitioner's formulation: the telephone of the 1970s did not have a thousand engineers on the other side redesigning it every day to make it more persuasive (CBS, 2017, a former Google product manager)—the numerator of the ratio has organized engineering doing work on it, and the denominator is a single host's finite, constant audit capacity—no engineering team works to expand it.
- **From Dashboard to Evolutionary Framework**: The symbiosis-parasitism continuum explains the long-term upward pressure on the cognitive Reynolds number as a directional indicator of the migration of control. It provides an evolutionary taxonomy basis for the domain projections of subsequent chapters (financial markets, AI history, and the confluence of the two).

A long-term investor who does not watch the tape or stare at live quotes is reducing the frequency of market readings received, decoupling personal meta-audit pacing from the pull of high-frequency readings, and reserving time for independent checking—what changes here is personal media exposure and auditing arrangements, not the governance clock defined in Chapter 3 as a complete institutional response cycle; the frequency of market quotations and the techno-capital clock are compared here only as temporal scales. The cognitive Reynolds number likewise offers direction rather than a precise threshold. Read (1898/1914) supplies a methodological note:

> It is better to be vaguely right than exactly wrong. (Read, 1898/1914)

In the language of Chapter 1, this is to read the clocks in the room as clocks again—acknowledging them as merely one source of readings, not the sun outside the room against which everything might be calibrated.

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§7 Falsifiability Declaration

This chapter distinguishes counterevidence to its core mechanism from conditions requiring revision of parameters or scope. The former must challenge the joint sequence from arrival pressure to checking work, unchecked commitments, and subsequent invocation; the latter require only updated reference magnitudes or revised classification boundaries and must not be presented as refutations.

The following two items concern the revision of parameters and scope:

- If future research revises the order-of-magnitude summary of human conscious throughput upward beyond the task-estimate range of Table 1 of the review (upper end about 50 bits/s), or shifts the order-of-magnitude summary itself by a full order of magnitude (the 10^2 bits/s range), then this reference magnitude and its conditions of applicability will need revision.
- If a media technology emerges that, while increasing the media conductivity coefficient, also embeds an effective automatic audit mechanism (such as a real-time fact-checking system proven to systematically shrink the fiber), then the scope of the cognitive Reynolds number's long-term upward tendency will need revision.

The core mechanism must be tested against joint outcomes rather than treating an improvement on the defense side as a refutation: if, under comparable hosts, tasks, accounting periods, and information units, and with other projection-side and defense-side conditions controlled, a sustained increase in the media conductivity coefficient does not compress the verification window, control over reception, or the trigger for checking, and unchecked commitments and their subsequent invocations do not increase, then the causal chain from arrival pressure to constrained auditing conditions and debt accumulation must be re-examined.

The result of McDonnell and Strayer (2024) is boundary evidence for the reversibility of meta-audit capacity, not a refutation of the framework, because §2.1 already includes state-dependent, reversible variation in that capacity. The mechanism claim requires revision only if such interventions, after exposure and task conditions are controlled, durably and systematically eliminate verification shortfalls and debt accumulation under high arrival pressure. The duration of the effect and the costs of recovery have no data as yet, and "highly dissipative" is not asserted as a proven claim; this change is not recorded as a restoration of the human biological substrate (E_c) itself.

The falsifiability conditions above can be tested through an equal-time budget protocol. The methodology of this protocol makes explicit the standing approach of this chapter: **form shapes the conditions**—the medium's form (mode of arrival, control over reception, trigger for checking, personalization precision) shapes the links at which cognitive pressure is generated and at which checking fails; it does not determine the specific content consequences. Statistical claims are confined to the specified default configurations and their periods of applicability; when a configuration changes, it is estimated as a new scenario—a single content counterexample does not refute a statistical tendency, but systematic contrary results under default usage conditions do constitute counterevidence.

The test takes one hour as the unified measurement window (the ratio depends only on arrival rate and reading speed and is independent of window length, §2.3). The protocol does not presuppose equal incoming workloads across media: books are recorded by the text actually entering the task, television by the content it broadcasts, and generative AI by the generated output entering the processing scope (§4's worked example: up to sev-

eral hundred thousand characters per hour—machine-generated volume is not capped by human consumption rates); what is actually read, watched, or understood is recorded separately and does not replace the arrival ledger (§2.2's ideal-type criterion). What distinguishes media is not the host's real-time consumption volume but arrival volume and debt structure. Memory encoding formed through visual reading is subject to a capacity constraint: visual short-term memory capacity (but neither maintenance duration nor the test process) predicts the effective encoding rate of visual long-term memory (Fukuda & Vogel, 2019). That finding concerns visual memory and prescribes no uniform formation rate for all commitments; generated content adopted without being read, and actions executed directly by agents, do not pass through this encoding channel. The protocol must therefore record, for each commitment individually, how it arose, whether it passed through reading, whether it was verified, and whether it entered subsequent invocation—this provides a partial mechanistic basis for per-commitment recording of generation times, with the extrapolation boundary as stated. The protocol records the cognitive commitments newly generated per hour—propositions accepted as true, citations written in, function behaviors adopted—and their destinations: how many are checked on the spot—among those verified, the ones that hold are recorded as settled, while ones where an error is found but correction or withdrawal is not yet complete remain outstanding debt (repair must include the necessary dependency repairs); how many are retained, together with their unchecked status, in a to-be-invoked state (active debt—the individual form of shadow debt); how many cease to enter any output or invocation within a prespecified follow-up period and have no identified downstream dependency, and are booked as an operational write-off (forgetting as such does not constitute a write-off); and those whose disposition cannot be determined are recorded as pending. For the retained portion, the protocol additionally records the independent verification and repair labor-time required for future settlement. Beyond the totals and destinations within the window, the protocol must record, for each commitment individually, the times of its generation, verification, invocation, and correction (distinguishing the time the output was generated from the time the host's commitment was formed), and track invocations and repairs after the window—only such time-series records can distinguish synchronous checking, steady accumulation, and dependency-chain propagation. The protocol records separately: the information arriving and requiring processing within the window and the verification labor each item requires (the numerator of the ratio); the host's calibration capacity (the basis for estimating the denominator, §2.2–§2.3; the measurement window is one hour, and the denominator is the length of the arrival stream itself); actual verification time and actual investment (behavioral records, not used to back out capacity); and commitments accepted and retained without verification, together with their subsequent invocations (the debt ledger, not derived automatically from arrival volume). A ratio persistently above 1, with unverified commitments becoming premises of subsequent output, is the accounting manifestation of the turbulent-type debt trajectory described below.

The objects of the laminar/transition/turbulent judgments thereby shift from the medium's default usage mode to the debt trajectory: commitments and checks proceeding largely in step mark the laminar type; arrears accumulating linearly, partly digestible through pauses and re-examination, mark the transition type; and the turbulent type is marked by some point after which unchecked commitments become premises for subsequent output, errors begin to propagate in chains, and repair requires rolling back an entire dependency chain. The protocol presupposes no ordering—an authoritatively written but factually unsound book can accumulate high debt, and adequately tested AI coding can correct errors on the spot without sedimentation—but directional evidence already exists for individual dimensions (Chiossi et al., 2023 found in a 60-participant between-subjects experiment that participants who used TikTok during the break showed significantly degraded prospective-memory performance on resumption, with no such effect under Twitter, YouTube, or rest—the combination of short

videos and rapid context switching impairs intention recall and execution; Su et al., 2021 found that personalized videos selectively activate the ventral tegmental area, a key node of the reward and addiction circuitry, compatible in direction with "personalization lengthens dwell time"); what is being tested is the set of conditions under which each formal dimension holds, not the proposition that media make no difference.

The equal-time protocol tests precisely this chapter's founding question: under what conditions a Fermi-scale directional judgment holds. The cognitive Reynolds number remains a clock that must be read as a clock—not the sun outside the room.

Conclusion

The cognitive Reynolds number is defensible not as a unified physical threshold across media, but as a discipline of scenario-specific Fermi estimation: a common measurement window aligns the workload awaiting audit with the normalization baseline, while ordinary reading, verification, actual investment, and unchecked commitments are recorded separately. Differences among books, television, short videos, and AI are shaped jointly by control over arrival, the verification window, the trigger for checking, and personalization precision; every numerical estimate holds only within its declared parameters and protocol.

Information-arrival pressure does not automatically equal cognitive debt. Pressure becomes a debt trajectory only when unchecked commitments are accepted, retained, and subsequently invoked, propagating through dependency chains. The three tasks named by the chapter title must therefore remain distinct: arrival pressure records how much work is required, verification capacity records what the full window and the ex ante budget permit, and debt trajectories record how unfinished yet invoked commitments propagate.

The cognitive Reynolds number remains a clock inside the room, not the sun outside it. It marks fibers and auditing gaps, while the Macro-Cognitive Accounting Framework records their consequences over time. Its value lies in making pressure, capacity, and debt separately visible, not in declaring reality precisely measured.

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